From: Institute Of Art and Design Jamaica (IoAaDJ)

IMMERSIVE DIGITAL MEDIA PRODUCTION

(Non-Photorealistic Rendering [NPR])
(Physically Based Rendering [PBR])
(Utopia-Edition)



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(PRD00) Production - "Immersive-Digital-Media-Production-I"

Programme	: Media Production
Department	: Media
Prerequisites	: CSEC® (Literature, Technical Drawing, VisualArts, TheatreArts, or Music) (CAPE® Performing Arts U1 and/or U2) or (satisfactory demo-reel/interview)
Type Of Course	: Production
Course Title	: "Immersive-Digital-Media-Production-I"
Course Code	: PRD00
Credits	: 3
Year	: Year-One (2026)
Semester	: s1-spring, s2-summer, s3-fall
Duration	: 1 Semester, 15 weeks, 60 hours (4 hours per week),(Every Wednesday@12:00am) (T1: January-01_May-01)(T2: June-01_August-31)(T3: September-01_Demember-01)
Lecturer / Tutor	: Israel Andrew Brown
Email	: israelandrewbrown@proton.me
Email (Organization)	: ioaadj@proton.me
Website	: https://github.com/israelandrewbrown/Immersive-Digital-Media-Production
Discord	: https://discord.gg/TsZG7jTQDA
Jitsi (Online-Meet)	: https://meet.proton.me/join/id-QF8T7C2RVR#pwd-oZaELJBJuukm
e-Resources	: thirty-embedded-video-links-twenty-hours-cited-throughout-the-document

Rationale :

This course is essential for bridging the gap between theoretical media studies and practical, industry-standard immersive digital content creation. By focusing extensively on the pipeline for traditionally-hand-drawn pre-production, the course ensures students are fully equipped to conceptualize, plan, and pre-visualize complex digital media projects. It fosters critical problem-solving skills and technical proficiency, laying the necessary groundwork for executing high-fidelity project-based productions in subsequent phases.

Description :

This course provides a comprehensive exploration of the immersive digital media pre-production workflow. Students will systematically journey through the complete pipeline, beginning with structured research and narrative development (treatment and screenplay), advancing into comprehensive visual development (character, prop, and environment [world] concept art), and culminating in meticulous audio (vo, sfx, music) planning and animatic workflows. Utilizing both 2D/3d storyboarding techniques and modern 2d/3D pre/post-visualization tools, learners will construct a robust, production-ready pre/post-production package.

Objectives :

1. Formulate comprehensive research and narrative treatments that effectively establish the foundation and scope for an immersive digital media project.
2. Design and document detailed visual concepts—including characters, props, and environments—while maintaining stylistic consistency and functional design.
3. Construct synchronized pre-visualization assets, including story-beats, lighting diagrams, and storyboards, to map out scene pacing and visual flow.
4. Integrate audio elements and visual sequences into dynamic 2D and 3D animatics that serve as a concrete, actionable blueprint for final production.

Course Assessment : "(PRJ00) Project – Individual Project"

Equipment (Required) : Laptop (<16gb ram) (<256gb ssd), e-Mouse, Microphone, camera, <50mbps internet, room

cost-of-course/exam : $375.00 USD ($1.00usd to 160jmd)(fee-subject-to-change)(payable-via-bitcoin) .
.Donate Bitcoin : bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v (request-btc-address-for-tuition)

This course material is intended for educational use and is based on the author's current understanding of the subject matter. While the author has taken reasonable steps to ensure the accuracy of the information presented, the rapidly evolving nature of artificial intelligence and media production means that some details may become outdated. The information in this book is distributed on an "As Is" basis without warranty. While every precaution has been taken in the preparation of the book, the author shall not have any liability to any person or entity with respect to any loss or damage caused or alleged to be caused directly or indirectly by the instructions contained in this book or products described in it.

IoAaDJ_3d-Generalist__ByWeek_LiveDemontration
1. Introduction 2. Research 3. Treatment 4. Screenplay 5. Scene_Character 6. Scene_Prop 7. Scene_World 8. Storybeat[pencil,ink,kripto/flat,colour]/LightDiagram 9. Audio_VoiceOver 10. Audio_SoundDesign 11. Audio_Music 12. Preliminary_Animatics_2d 13. Preliminary_Storyboard_2d 14. Post_Animatics_3d 15. Post_Storyboard_3d

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StudioBinder. “Three Act Structure Explained — the Secret to Telling a Great Story.” *YouTube*, 24 July 2023, www.youtube.com/watch?v=tvqjp1CxxD8.

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Title-Of-Lesson	Introduction / Course Orientation 01:
Aim	:To introduce students to the complete digital media pre-production pipeline, establishing course expectations, software requirements, and the conceptual framework required for successful project development.
Objectives	: Identify the 14 core stages of the immersive production pipeline. Configure the required hardware and software environments. Understand the criteria and deliverables for the final pre-production package.
Required Materials	: Laptop, internet connection, syllabus, access to Discord and Jitsi.
Lecture	:This session provides a high-level overview of the entire course. We will discuss the evolution of media production pipelines, the critical importance of rigorous pre-production, and how each weekly phase builds upon the previous one to construct a comprehensive film package.
Tutorial	:Guided walkthrough of setting up the digital workspace. Students will join the class Discord, test their Jitsi connection, and review the open-source tools required for the semester.
Assignment	: Set up digital project folders and submit a brief written proposal outlining potential project ideas.
Due Date	:End of Week 1

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Title-Of-Lesson-02	Research
Aim	:To train students in foundational research methodologies tailored for narrative media, focusing on the 5 Ws to establish a strong contextual base for their projects.
Objectives	:Apply the 5 Ws and Bonus H to deconstruct narrative concepts. Formulate a structured abstract and methodology for the project. Compile initial visual and thematic references.
Required Materials	:Word processor, internet browser, research notebook.
Lecture	:Exploration of academic and practical research in media. We will break down how defining the When, Where, Who, Why, and What anchors a story in reality and informs design choices later in the pipeline.
Tutorial	:Interactive session where students workshop their project ideas using the 5Ws framework, receiving peer and instructor feedback to refine their project scope.
Assignment	: Deliver the Research-Paper/Presentation draft covering Abstract, Methodology, and References. And Presentation on who, what, where, when and why.
Due Date	:End of Week 2
Reference	:Khan Academy. “Using the 5 Ws and the Bonus H to Make Summaries | Reading | Khan Academy.” YouTube, 20 Feb. 2025, https://www.youtube.com/watch?v=T3LsopJ4dec. Accessed 26 Feb. 2025.

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Title-Of-Lesson-03	: Treatment
Aim	:To develop a concise and compelling narrative treatment that outlines the project's core story, characters, and structural flow.
Objectives	:Draft a compelling log-line that captures the essence of the project. Write a comprehensive synopsis utilizing the three-act structure. Define the primary protagonist and antagonist roles.
Required Materials	: Word processor, previous research document.
Lecture	:Detailed breakdown of the three-act structure and character archetypes. We will analyze how treatments act as the critical bridge between abstract research and formatted screenplays.
Tutorial	:Writing workshop focusing on crafting tight log-lines and expanding them into structured synopses. Students will evaluate each other's protagonist/antagonist dynamics.
Assignment	: Submit the complete project Treatment (Author, Title, Log-line, Synopsis, Characters).
Due Date	:End of Week 3

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Title-Of-Lesson-04	:Screenplay
Aim	:To translate the narrative treatment into a professionally formatted screenplay, mastering industry-standard script elements and dialogue formatting.
Objectives	:Understand the anatomy and strict formatting rules of a standard screenplay. Install and configure Trelby (or equivalent screenplay software). Draft the first 3 to 5 pages of the project script
Required Materials	: Screenwriting software (Trelby recommended), Treatment document.
Lecture	:Introduction to screenplay syntax—scene headings, action lines, character names, parentheticals, and dialogue. Discussion on why standardized formatting is crucial for the entire production crew.
Tutorial	:Hands-on software setup. Students will install Trelby, apply the custom UI theme if desired, and begin adapting their treatment into script format under instructor guidance.
Assignment.	:Submit a 3 to 15-page properly formatted Screenplay draft
Due Date	:End of Week 4

StudioBinder. “Three Act Structure Explained — the Secret to Telling a Great Story.” *YouTube*, 24 July 2023, www.youtube.com/watch?v=tvqjp1CxxD8.

StudioBinder. “What Is a Protagonist: A Breakdown of Different Types and Functions of the Main Character.” *YouTube*, 2 Jan. 2023, www.youtube.com/watch?v=pqQTv8KJxQc. Accessed 20 Feb. 2025.

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Title-Of-Lesson-05	: Scene_Character
Aim	:To initiate the visual development phase by designing compelling, production-ready concept art for the protagonist and antagonist.
Objectives	:Translate written character bios into visual designs. Produce character turnarounds (5 views or T-pose/A-pose). Illustrate expressive face charts and dynamic action poses.
Required Materials	: Digital drawing tablet/e-Mouse, 2D illustration software (e.g., Krita, Photoshop).
Lecture.	:Fundamentals of character design, including shape language, silhouette, and functional anatomy. We will examine how character design communicates personality and backstory without words
Tutorial	:Digital drafting session. Students will block out primary character silhouettes and begin drafting T-poses and expression sheets.
Assignment	: Submit Concept Art for two characters (protagonist and antagonist) including bios, turnarounds, expressions, and action poses.
Due Date	:End of Week 5
Reference	:Tyler Edlin. “DESIGN BETTER CHARACTERS: Essential Fundamentals.” YouTube, 2024, https://www.youtube.com/watch?v=9AgGCtfbuLs&ab_channel=TylerEdlin.

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Title-Of-Lesson-06	: Scene_Prop
Aim	:To design functional and thematic props that enhance the narrative and integrate seamlessly with the character designs.
Objectives	:Identify key narrative props from the screenplay. Design props with attention to scale, functionality, and aesthetic consistency. Create detailed orthographic views of essential props.
Required Materials	: Digital illustration software, Screenplay.
Lecture	:Discussion on the role of props in storytelling and world-building. We will explore industrial design principles as they apply to fantasy, sci-fi, and realistic digital props.
Tutorial	:Studio time dedicated to ideation sketching and finalizing prop orthographics. Emphasis on maintaining consistent scale relative to the character T-poses.
Assignment	:Submit Concept Art documentation for primary scene props.
Due Date	:End of Week 6
Reference	:Tyler Edlin. “Mastering the Design Pipeline.” YouTube, 2024, https://www.youtube.com/watch?v=_15SIWLGQEU&ab_channel=TylerEdlin

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Title-Of-Lesson-07	: Scene_World
Aim.	:To conceptualize and document the spatial environments of the project, establishing the setting through thoughtful composition and layout
Objectives	:Design environment concepts utilizing the hemisphere and set (cube) methodology. Establish clear spatial segmentation and camera-ready composition. Unify characters, props, and world aesthetics.
Required Materials	: Illustration/3D blockout software.
Lecture	:Principles of environment design: perspective, atmospheric depth, and focal points. We will discuss how to construct a digital set as a functional 'cube' for camera staging.
Tutorial	:Drafting environment thumbnails and refining one chosen environment into a polished concept piece showing segmentation and composition.
Assignment	: Submit Concept Art for the environment (hemisphere, set, segmentation, composition).
Due Date	:End of Week 7
Reference:	Tyler Edlin. “The BEST Environment Design EXERCISE for BEGINNERS.”YouTube, 2019, https://www.youtube.com/watch?v=mhvtuZIEV_Q&ab_channel=TylerEdlin.

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Title-Of-Lesson-08	: Storybeat/LightingDiagram
Aim	:To map the visual narrative pacing and establish the mood through keyframe story-beats and lighting diagrams.
Objectives	:Identify the critical narrative beats from the screenplay. Create colored images representing important events (keyframes). Draft lighting diagrams to establish scene mood and visibility.
Required Materials	: 2D software, concept art assets.
Lecture	:Introduction to cinematography and lighting theory. We will study shot sizes, camera angles, and how lighting direction impacts psychological perception of a scene.
Tutorial	:Color scripting and lighting blockouts. Students will overlay lighting diagrams onto their environment concepts to plan practical or digital light sources.
Assignment	: Submit Story-Beats/Lighting-Diagram (coloured images of important events/panels).
Due Date	:End of Week 8
Reference	

Proko. “How to Create Iconic Comic Covers.” YouTube, 10 Apr. 2025, www.youtube.com/watch?v=MxTKZfZ7WWU.

Bob Draws! “Understanding the Krita Comics Template.” YouTube, 6 Sept. 2022, www.youtube.com/watch?v=W8X0139fO2U. Accessed 24 Mar. 2025.

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Proko. “Coloring Spiderman with Copic Markers.” YouTube, 25 Aug. 2022, www.youtube.com/watch?v=2v-JLiepI0A. Accessed 11 Apr. 2026.

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Title-Of-Lesson-09	: Audio_VoiceOver
Aim	:To capture and process high-quality expressive voiceover recordings that will drive the pacing of the animatic.
Objectives	:Understand the science of sound recording in varied environments. Record expressive dialogue based on the screenplay. Clean and process audio tracks using Audacity.
Required Materials	: Microphone, Audacity software, quiet room.
Lecture	:The fundamentals of acoustics, microphone placement, and vocal performance. We will review common audio issues (clipping, background noise) and how to avoid them during recording.
Tutorial	:Practical recording session followed by an Audacity workshop. Students will learn noise reduction, equalization, and compression to improve voice clarity.
Assignment	:Submit processed VoiceOver audio tracks for the project.
Due Date	:End of Week 9:

StudioBinder. “The Science of Sound in Film — Film Sound Recording for Beginners.” *YouTube*, 29 July 2024, www.youtube.com/watch?v=mXtnHHJFREM.

Goobster's Room. “Every Major Audio Effect Explained in 8 Minutes!” *YouTube*, 16 Mar. 2025, www.youtube.com/watch?v=JGGnla8RKP4. Accessed 12 Apr. 2025.

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Title-Of-Lesson-10	: Audio_SoundDesign
Aim	:To develop a rich auditory landscape by integrating sound effects and foley that enhance the realism and impact of the visual narrative.
Objectives	:Source and organize a library of appropriate sound effects. Apply major audio effects (reverb, delay, pitch shift) creatively. Begin layering Foley with the VoiceOver tracks.
Required Materials	: Audio editing software, access to royalty-free SFX libraries.
Lecture	:Introduction to Sound Design theory. We will discuss how ambient noise, foley, and hard effects create immersion and guide the audience's attention.
Tutorial	:Audio mixing workshop. Students will layer environmental ambiences and action-specific sound effects to build a cohesive soundscape blockout.
Assignment	:Submit the initial Sound Design mix (SFX + VO).
Due Date	:End of Week 10

StudioBinder. “The Science of Sound in Film — Film Sound Recording for Beginners.” *YouTube*, 29 July 2024, www.youtube.com/watch?v=mXtnHHJFREM.

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Title-Of-Lesson-11	: Audio_Music
Aim.	:To finalize the project's soundscape by selecting, editing, and integrating musical scores that elevate the emotional tone of the animatic
Objectives	
Required Materials	:Select appropriate musical tracks that match the narrative mood. Edit music to match story-beats and transitions. Master the final audio mix (VO, SFX, Music) for the animatic.
Tutorial	
Assignment	: Audio editing software, royalty-free music tracks. Lecture: The psychological impact of music in the media. Discussion on pacing, rhythm, and how to mix music so it supports rather than overpowers dialogue and sound effects.
Due Date	
	:Final audio balancing. Students will adjust levels, pan audio for spatial awareness, and prepare the master audio file for the animatic phase. (Subtitles overview utilizing Aegisub).
	:Submit the finalized Master Audio Track.
	:End of Week 11

StudioBinder. “The Science of Sound in Film — Film Sound Recording for Beginners.” *YouTube*, 29 July 2024, www.youtube.com/watch?v=mXtnHHJFREM.

Goobster's Room. “Every Major Audio Effect Explained in 8 Minutes!” *YouTube*, 16 Mar. 2025, www.youtube.com/watch?v=JGGnla8RKP4. Accessed 12 Apr. 2025.

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Title-Of-Lesson-12	: Preliminary_Storyboard_2d
Aim	:To visually map the screenplay into sequential 2D storyboard panels, clearly indicating action, character placement, and camera movement.
Objectives	:Translate script scenes into sequential visual panels. Utilize standard storyboard notation (arrows, color coding). Establish screen direction and adhere to the 180-degree rule.
Required Materials	: Storyboarder software, digital drawing tools.
Lecture	:Fundamentals of visual storytelling through sequential art. We will review how to represent complex camera movements (pans, tilts, tracking) within static 2D panels.
Tutorial	:Introduction to the Storyboarder application. Students will begin drawing out their scenes, applying color codes to characters and clear directorial arrows.
Assignment	:Submit Preliminary 2D Storyboard PDF/.cbz.
Due Date	:End of Week 12

StudioBinder. “Ultimate Guide to Camera Shots: Every Shot Size Explained.” *YouTube*, 4 May 2020, www.youtube.com/watch?v=AyML8xuKfoc.

StudioBinder. “Directing Camera Movement Filmmaking Techniques for Directors: .” *YouTube*, 30 July 2018, www.youtube.com/watch?v=GbnYBmqBbKA.

John McCaffrey. “Getting Started with Storyboarder.” *YouTube*, 9 Feb. 2021, www.youtube.com/watch?v=dUVaubWgkD0. Accessed 15 Dec. 2025.

John McCaffrey. “Using Storyboarder to Create Animatics.” *YouTube*, 21 Apr. 2020, www.youtube.com/watch?v=huqDOlSsyml. Accessed 15 Dec. 2025.

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Title-Of-Lesson-13	: Preliminary_Animatics_2d
Aim	:To synchronize the 2D storyboard panels with the master audio track, creating a timed video animatic that pre-visualizes the final product.
Objectives	:Import 2D storyboard panels into an editing timeline. Sync panels to the VoiceOver, SFX, and Music track. Evaluate and adjust pacing based on audio cues.
Required Materials	: Storyboarder / Video Editing Software, Master Audio file.
Lecture:	The art of the animatic. Discussion on rhythm, timing, and how editing dictates the audience's perception of time and emotion in digital media.
Tutorial	:Editing lab. Students will use Storyboarder’s timeline features (or equivalent NLE) to cut their panels to the beat of their audio, applying basic panning/zooming effects to panels.
Assignment	:Submit Preliminary 2D Animatic video file.
Due Date	:End of Week 13

StudioBinder. “Ultimate Guide to Camera Shots: Every Shot Size Explained.” *YouTube*, 4 May 2020, www.youtube.com/watch?v=AyML8xuKfoc.

StudioBinder. “Directing Camera Movement Filmmaking Techniques for Directors: .” *YouTube*, 30 July 2018, www.youtube.com/watch?v=GbnYBmqBbKA.

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John McCaffrey. “Using Storyboarder to Create Animatics.” *YouTube*, 21 Apr. 2020, www.youtube.com/watch?v=huqDOlSsymI. Accessed 15 Dec. 2025.

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Title-Of-Lesson-14	: Post_Animatics_3d
Aim	:To transition from 2D planning to 3D pre-visualization, utilizing advanced shot management tools to block out accurate camera data and character placement.
Objectives	:Set up a 3D workspace (Blender) for previs workflows. Utilize Shot Manager/Storyliner addons to configure cameras. Block out key scenes in a 3D spatial environment.
Required Materials	: Blender, Storyliner / Ubisoft Shotmanager addons.
Lecture	:Introduction to modern 3D pre-production pipelines. We will explore how tools like Ubisoft's Shot Manager bridge the gap between storyboards and final layout, saving massive production time.
Tutorial	:Technical lab focused on addon installation and basic 3D blocking. Students will import rough proxy geometry based on their Week 7 environments and set up their cameras.
Assignment	:Submit 3D Animatic blockout files.
Due Date	:End of Week 14

StudioBinder. “Ultimate Guide to Camera Shots: Every Shot Size Explained.” *YouTube*, 4 May 2020, www.youtube.com/watch?v=AyML8xuKfoc.

StudioBinder. “Directing Camera Movement Filmmaking Techniques for Directors: .” *YouTube*, 30 July 2018, www.youtube.com/watch?v=GbnYBmqBbKA.

WerwackFx. “Werwack/Storyliner-Support” *GitHub*, 12 June 2025, www.github.com/werwack/storyliner-support.

Spitfire Storyboards. “STORYLINER - Features Overview of a Truly Phenomenal Addon.” *YouTube*, 31 Dec. 2024, www.youtube.com/watch?v=Q-RI5MaCctQ.

ubisoft. “GitHub - Ubisoft/Shotmanager: Shot Manager.” *GitHub*, 27 Feb. 2023, www.github.com/ubisoft/shotmanager.

Spitfire Storyboards. “‘ubisoft shot manager’ (now ‘storyliner’) (previs, hybrid + storyboard tools).” *YouTube,* 16 Oct. 2022, https://www.youtube.com/watch?v=8yBjm6ojwe8.

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Title-Of-Lesson-15	: Post_Storyboard_3d / Final Integration
Aim	:To compile, finalize, and package all pre-production assets into a cohesive, professional document (.pdf or .cbz) for final assessment.
Objectives	: Extract final 3D storyboard frames from the previs setup. Assemble research, scripts, concept art, and boards into a master layout. Review all deliverables against the course grading rubric.
Required Materials	: All generated course materials, PDF/CBZ compiler software.
Lecture	:The importance of presentation. Discussion on how a well-organized pre-production package is pitched to stakeholders, studios, or utilized efficiently by a large production team.
Tutorial	:Compilation and formatting workshop. Students will finalize their Title Cards, ensure consistent formatting across all documents, and export their final packages.
Assignment	: Submit the Complete Film Pre-production Package (.pdf or .cbz) (Course Assessment PRJ00).
Due Date	: End of Week 15

StudioBinder. “Ultimate Guide to Camera Shots: Every Shot Size Explained.” *YouTube*, 4 May 2020, www.youtube.com/watch?v=AyML8xuKfoc.

StudioBinder. “Directing Camera Movement Filmmaking Techniques for Directors: .” *YouTube*, 30 July 2018, www.youtube.com/watch?v=GbnYBmqBbKA.

WerwackFx. “Werwack/Storyliner-Support” *GitHub*, 12 June 2025, www.github.com/werwack/storyliner-support.

Spitfire Storyboards. “STORYLINER - Features Overview of a Truly Phenomenal Addon.” *YouTube*, 31 Dec. 2024, www.youtube.com/watch?v=Q-RI5MaCctQ.

ubisoft. “GitHub - Ubisoft/Shotmanager: Shot Manager.” *GitHub*, 27 Feb. 2023, www.github.com/ubisoft/shotmanager.

Spitfire Storyboards. “‘ubisoft shot manager’ (now ‘storyliner’) (previs, hybrid + storyboard tools).” *YouTube,* 16 Oct. 2022, https://www.youtube.com/watch?v=8yBjm6ojwe8.

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(PRD00) Production - "Immersive-Digital-Media-Production-IIa"

Programme	: Media Production
Department	: Media
Prerequisites	: CSEC® (Literature, Technical Drawing, VisualArts, TheatreArts, or Music) (CAPE® Performing Arts U1 and/or U2) or (satisfactory demo-reel/interview)
Type Of Course	: Production
Course Title	: "Immersive-Digital-Media-Production-II"
Course Code	: PRD00
Credits	: 3
Year	: Year-One (2026)
Semester	: s1-spring, s2-summer, s3-fall
Duration	: 1 Semester, 15 weeks, 60 hours (4 hours per week),(Every Wednesday@12:00am) (T1: January-01_May-01)(T2: June-01_August-31)(T3: September-01_Demember-01)
Lecturer / Tutor	: Israel Andrew Brown
Email	: israelandrewbrown@proton.me
Email (Organization)	: ioaadj@proton.me
Website	: https://github.com/israelandrewbrown/Immersive-Digital-Media-Production
Discord	: https://discord.gg/TsZG7jTQDA
Jitsi (Online-Meet)	: https://meet.proton.me/join/id-QF8T7C2RVR#pwd-oZaELJBJuukm
e-Resources	: thirty-embedded-video-links-twenty-hours-cited-throughout-the-document

Rationale :

Control constitutes a fundamental principle underlying all aspects of 3D production. Whether forming, texturing, rigging, moving, a character, prop, world, elements or integrating CGI into live action footage, the artist's role is to form, colour, control, move, illuminate, and perceive; and utilize available standard control tools. This course isolates that idea—training students to deliberately manipulate systems instead of relying on automation or presets.

Description :

This course examines control as both a conceptual framework and a technical discipline. Emphasis is placed on the deliberate manipulation of computational systems, tools, and workflows that govern digital content creation. Students will engage with core methodologies that enable precise parameterization, procedural adjustment, and predictable output across production stages. By isolating and interrogating the notion of control, the course prioritizes a deep operational understanding of underlying systems over reliance on automated processes or predefined presets. Through applied exercises and critical analysis, students will develop the capacity to construct, modify, and evaluate controlled outcomes within complex 3D environments. The course thereby establishes a rigorous foundation for advanced practice, fostering technical fluency, analytical thinking, and intentional authorship in digital production contexts.

Objectives :

1. Develop a rigorous understanding of how control underpins modeling, texturing, rigging, animation, lighting, and compositing in 3D production.
2. Acquire technical proficiency in manipulating tools, parameters, and procedural systems to achieve predictable and intentional results, rather than relying on automation or presets.
3. Apply control methodologies across interconnected production stages, including CGI integration into live-action footage, ensuring consistency and precision throughout the pipeline.
4. Analyze and assess production processes, refining workflows to enhance efficiency, visual quality, and creative intent in complex 3D environments.

Course Assessment : "(PRJ00) Project – Individual Project"

Equipment (Required) : Laptop (<16gb ram) (<256gb ssd), e-Mouse, Microphone, camera, <50mbps internet, room

cost-of-course/exam : $375.00 USD ($1.00usd to 160jmd)(fee-subject-to-change)(payable-via-bitcoin) .

.Donate Bitcoin : bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v (request-btc-address-for-tuition)

This course material is intended for educational use and is based on the author's current understanding of the subject matter. While the author has taken reasonable steps to ensure the accuracy of the information presented, the rapidly evolving nature of artificial intelligence and media production means that some details may become outdated. The information in this book is distributed on an "As Is" basis without warranty. While every precaution has been taken in the preparation of the book, the author shall not have any liability to any person or entity with respect to any loss or damage caused or alleged to be caused directly or indirectly by the instructions contained in this book or products described in it.

IoAaDJ_3d-Generalist__ByWeek_LiveDemontration

1. Introduction
2. VFX_IntersectingCylinders
3. VFX_Sculpt_head
4. VFX_nodes/layers_PBR_Cylinder_SprayCan
5. VFX_rig&picker_Ball, BallTail, BallPendulum
6. VFX_rig&picker_Humaniod_Cylinder_Body
7. VFX_rig&picker_Humaniod_Ball_Face
8. VFX_keyframes_Ball, BallTail, BallPendulum
9. VFX_keyframes_Humaniod
10. VFX_keyframes_Humaniod_mixed
11. VFX_MattePainting
12. VFX_CameraTrack
13. VFX_Rotoscope
14. VFX_ChromaKey
15. VFX_PanoramicStitched_HDR/PBR

Checklist_IoAaDJ_3d_Generalist_Portfilio_ImagesVideos

1. [ ] VFX_IntersectingCylinders
2. [ ] VFX_Sculpt_head
3. [ ] VFX_nodes/layers_PBR_Cylinder_SprayCan
4. [ ] VFX_rig&picker_Ball
5. [ ] VFX_rig&picker_BallTail
6. [ ] VFX_rig&picker_BallPendulum
7. [ ] VFX_rig&picker_Humaniod_CylinderBody
8. [ ] VFX_rig&picker_Humaniod_ball_Face
9. [ ] VFX_keyframes_Ball
10. [ ] VFX_keyframes_BallTail
11. [ ] VFX_keyframes_BallPendulum
12. [ ] VFX_keyframes_Humaniod_Body
13. [ ] VFX_keyframes_Humaniod_Face
14. [ ] VFX_keyframes_Humaniod_mixed
15. [ ] VFX_CameraTrack
16. [ ] VFX_Rotoscope
17. [ ] VFX_ChromaKey
18. [ ] VFX_HDR/PBR
19. [ ] VFX_MattePainting
20. [ ] VFX_Light

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Lecture
Digitalist. “Every 3D Concept Explained (Basics).” *YouTube*, 15 Sept. 2025, www.youtube.com/watch?v=p3MU7J-0284. Accessed 26 Nov. 2025. Digitalist. “3D Modelling Explained (Basics).” *YouTube*, 6 Oct. 2025, www.youtube.com/watch?v=bChytx8PHvU. Digitalist. “3D Materials Explained (Basics).” *YouTube*, 19 Oct. 2025, www.youtube.com/watch?v=abo6ikU-Lx8. Accessed 26 Nov. 2025. Digitalist. “UV Unwrapping Explained (Basics).” *YouTube*, 28 Oct. 2025, www.youtube.com/watch?v=gyAUPI_PMRQ. Accessed 26 Nov. 2025 Digitalist. “Every Rigging Concept Explained, in 3D.” *YouTube*, 15 Dec. 2025, www.youtube.com/watch?v=1ufeIPub3B0. Accessed 16 Dec. 2025. Digitalist. “3D Animation Explained (Basics).” *YouTube*, 10 Nov. 2025, www.youtube.com/watch?v=keKNzdYiaBc. Accessed 26 Nov. 2025. Vox. “Why Visual Effects Artists Love This Shiny Ball.” *Youtube*, www.youtube.com/watch?v=HCfHQL4kLnw.

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Tutorials (Free-and-Open-Source) (15-hours)
Blender Bob. “Intersecting Cylinders in Blender.” YouTube, 24 Aug. 2023, www.youtube.com/watch?v=79d-QFMxQm8. Accessed 17 Mar. 2026. Ryan King Art. “Understanding Parenting in Blender.” *YouTube*, 22 Dec. 2023, www.youtube.com/watch?v=5oclP4hQndc. Accessed 22 Nov. 2024. FredericCervini. “Maya to Blender Tips: How to Move Your Pivot Point (Origin Point) like in Maya.” YouTube, 15 June 2022, www.youtube.com/watch?v=7hCHlaf7t5k. Accessed 20 Feb. 2026. My CG Tutor. “How to Freeze Transforms in Blender (like Maya).” *YouTube*, 19 Dec. 2024, www.youtube.com/watch?v=jo_myEEIDt0. Accessed 15 Dec. 2025.
CG Boost. “Sculpting in Blender for Beginners (Full Course/Tutorial) #B3d.” YouTube, 18 July 2024, www.youtube.com/watch?v=KURuPAVJ6hM. Abe Leal 3D. “Blender Hair System: Step-By-Step Tutorial.” YouTube, 31 July 2023, www.youtube.com/watch?v=rd3RkBVJMQA.
Ryan King Art. “Ucupaint for Beginners! (Free Blender Texture Painting Addon) 🖌️.” YouTube, 2 Apr. 2026, www.youtube.com/watch?v=i89WJ-ap4C4. Accessed 7 Apr. 2026.
Carlino, Joey. “Rigging for Impatient People - Blender Tutorial.” *Youtube*, 5 May 2023, www.youtube.com/watch?v=DDeB4tDVCGY. Matthew DeSa Art. “Rigging Ball and Tail in Blender Livestream.” *YouTube*, 12 Mar. 2024, www.youtube.com/watch?v=nNEx36i5jI8. Accessed 29 Nov. 2025. Wee learn from Mobile. “How to Model & Rig Simple Pendulum in Blender.” *YouTube*, 13 May 2023, www.youtube.com/watch?v=1eCQUI4Smco. CGDive (Blender Rigging Tuts). “What Makes This the ULTIMATE Ball Rig for Blender?” YouTube, 23 Mar. 2026, www.youtube.com/watch?v=HyD7dI0J_bA. Accessed 15 June 2026. CGDive (Blender Rigging Tuts). “BEGINNER Blender Rigging Course (2026).” YouTube, 8 Apr. 2026, www.youtube.com/watch?v=Ldki5fxSn5U. Accessed 8 Apr. 2026. CGDive (Blender Rigging Tuts). “NOVICE Blender Rigging Course (2026) [Reupload].” YouTube, 19 May 2026, www.youtube.com/watch?v=FzQVpzXcFX0. Accessed 15 June 2026. CGDive (Blender Rigging Tuts). “University Level Rigging Course for Free!” YouTube, 26 May 2026, www.youtube.com/watch?v=A147p_zDwO4. Accessed 15 June 2026. CGDive (Blender Rigging Tuts). “EASY Rigging with Blender Rigify: A Full Course.” YouTube, 1 June 2026, www.youtube.com/watch?v=FhF54DZY34U. Accessed 15 June 2026. Pierrick Picaut. “Dynamic Cloth Rigging and Animation in Blender.” YouTube, 14 Oct. 2022, www.youtube.com/watch?v=kCn4sProaek.
richstubbsanimation. “Create Dynamic Muscle Rigs in Blender 4.0.” YouTube, 17 Nov. 2023, www.youtube.com/watch?v=mnQhzwCipQA. Accessed 15 June 2026.
Alex on Story. “How to Animate in Blender: Learning the Basics.” *YouTube*, 1 Mar. 2024, www.youtube.com/watch?v=sM4_rbrQMpo. Accessed 22 Dec. 2025. Alex on Story. “How to Animate in Blender: Learning the Basics	Part 2.” *YouTube*, 13 Mar. 2024, www.youtube.com/watch?v=34IfSEusWCA. Accessed 29 Nov. 2025. Alex on Story. “How to Animate in Blender: Learning the Basics	Part 3.” *YouTube*, 8 Apr. 2024, www.youtube.com/watch?v=ENnPtjT5mBs. Accessed 22 Dec. 2025. Alex on Story. “Mastering the Blender Walk Cycle: Tips and Tricks for Animators!” *Youtube*. www.youtube.com/watch?v=MzJZ7yEEgRA. Accessed 26 Apr. 2023. Alex on Story. “Pose to Pose : Blender Animation Workflow for Beginners.” *Youtube*, 16 July 2023, www.youtube.com/watch?v=p8Bi7k60IS0.
Jacob Zirkle. “The Best Method for Camera Tracking in Blender 5.0.” *YouTube*, 4 Nov. 2025, www.youtube.com/watch?v=Jdc3hHHOdnk. Accessed 8 Dec. 2025. Jacob Zirkle. “How to Remove Tracking Markers in Blender.” *YouTube*, 22 Sept. 2023, www.youtube.com/watch?v=dPO2SQWV5_0. Accessed 10 Dec. 2025. Jacob Zirkle. “Learn Green Screen VFX in Blender in under 7 MINUTES.” *YouTube*, www.youtube.com/watch?v=t2WlJlXDNjA. Thies Grünewald. “VFX with Blender and Beeble Switchlight 2.0.” *YouTube*, 29 July 2025, www.youtube.com/watch?v=6fphiju5HKg. Accessed 26 Nov. 2025. InLightVFX. “Finally! A VFX Workflow with Less Guesswork.” *YouTube*, 1 Feb. 2023, www.youtube.com/watch?v=G2SacOKhJto. TechSimplify. “How to Adjust Exposure on IPhone Camera?” *YouTube*, 15 Sept. 2024, www.youtube.com/watch?v=3mO6wWzvcMI. Creative Pad Media. “GIMP HDR Tutorial - How to Merge Different Exposures with One Click.” *YouTube*, 22 Nov. 2023, www.youtube.com/watch?v=Ot7iGhiam6U. Accessed 26 Nov. 2025. Alexander Kiryanov. “Meshroom. Hugin. Create 360 Degree HDRI Panorama (Free PTGui Alternative).” *YouTube*, 22 Nov. 2020, www.youtube.com/watch?v=iufXTCPln_Q. Accessed 2 Jan. 2026.
Spitfire Storyboards. “‘ubisoft shot manager’ (now ‘storyliner’) (previs, hybrid + storyboard tools).” *YouTube,* 16 Oct. 2022, https://www.youtube.com/watch?v=8yBjm6ojwe8. Pierrick Picaut. “Master Production Workflow in Blender!” *YouTube*, 18 Feb. 2026, www.youtube.com/watch?v=JckDEbIB7qc. Accessed 20 Feb. 2026. Pierrick Picaut. “Everything about Linking & Overrides in Blender.” *YouTube*, 27 Jan. 2026, www.youtube.com/watch?v=z-KH-zmV79I. Accessed 20 Feb. 2026. Pierrick Picaut. “Shot, Scene, Sequence Explained Simply.” *YouTube*, 12 Jan. 2026, www.youtube.com/watch?v=lHqKbB6_Nqc. Accessed 20 Feb. 2026. Pierrick Picaut. Render Multiple Files, Scenes & Layers at Once in Blender Cycles + EEVEE. Video, YouTube, 2 Oct. 2026, https://www.youtube.com/watch?v=gDFy2hlWDLk. Accessed 6 Oct. 2026. ExplainingComputers. “Blender Video Editing: Learn Workflow, Transitions, Modifiers, More!” *YouTube*, 14 Sept. 2025, www.youtube.com/watch?v=99ZdOuJMrao. Accessed 26 Feb. 2026

Tutorials (Closed-Source-and-Proprietary) (15-Hours)
JL Mussi. “Maya Modeling Tutorial: 3d Modeling Cylindrical Corners & Intersections.” YouTube, 28 Jan. 2021, www.youtube.com/watch?v=ercSV_U66gU. Accessed 11 Apr. 2026. jc Tools. “Maya Tutorial: Everything on How to Move the Pivot Point in Maya.” YouTube, 15 July 2020, www.youtube.com/watch?v=rli51Rli8Tgs. Accessed 4 Apr. 2026. AnimSchool. “All about Parenting 3D Objects in Maya.” YouTube, 17 Mar. 2021, www.youtube.com/watch?v=1aDz51RWT2A. Accessed 4 Apr. 2026. RBGore! “Maya - Why Freeze Transforms in Maya??” YouTube, 10 Jan. 2025, www.youtube.com/watch?v=tox2GqShdys. Accessed 10 Apr. 2026. What Make Art. “Mastering Maya: Importing vs Referencing Scene Files - a Comprehensive Tutorial.” YouTube, 2 Apr. 2021, www.youtube.com/watch?v=WDpYKvhSOYI. Accessed 2 May 2026.
FlippedNormals. “Getting Started with Sculpting - ZBrush for Beginners Tutorial.” YouTube, 28 Feb. 2018, www.youtube.com/watch?v=_yKGfcp2z3k. FlippedNormals. “Retopology for Beginners in Maya.” YouTube, 29 Jan. 2018, www.youtube.com/watch?v=xnDWfaBQ3n8. Hill, J. “How to Make DOPE CG HAIR	XGen Tutorial.” Www.youtube.com, 16 Dec. 2021, www.youtube.com/watch?v=hvNcRaPmWoE.
FlippedNormals. “Introduction to UV Mapping - Learn the Complete Basics.” YouTube, 2 Aug. 2018, www.youtube.com/watch?v=diUuXid9oGo. J Hill. “How to TEXTURE EVERYTHING in Substance Painter.” YouTube, 8 Apr. 2026, www.youtube.com/watch?v=GvifkhCW3aM. Accessed 10 Apr. 2026. J Hill. “How to Make SKIN TEXTURES in Substance Painter - YouTube.” Youtube, 12 November 2021. www.youtube.com/watch?v=s0DhvFML7oM. Accessed 10 Apr. 2026. FlippedNormals Marketplace. “Quick Guide to Texture Baking in Substance Painter.” YouTube, 29 Apr. 2021, www.youtube.com/watch?v=cogWpYQ4WVQ.
Dikko. “Create an Advanced Ball Rig in 30 Minutes with Maya!” YouTube, 5 Oct. 2022, www.youtube.com/watch?v=MjvjrURL-Dg. Accessed 11 Apr. 2026. Stu Lloyd (CG Stu). “Rig a Ball in Maya (with Squash & Stretch, Tail).” YouTube, 11 May 2023, www.youtube.com/watch?v=gCLwluNwF0c. GMS VIDEOS. “Pendulum Rigging with P-Parent, Constrain, Joint & Bind Skin Methods in Autodesk Maya 2017.” YouTube, 25 Jan. 2020, www.youtube.com/watch?v=8QMEmhO-ANg. Accessed 11 Apr. 2026. ngSkinTools. “NgSkinTools2: Skinning Character Body.” YouTube, 16 June 2022, www.youtube.com/watch?v=xAi6cSyAV9M. Accessed 22 Mar. 2026.
SARKAMARI. “Character Rigging for Beginners: 01 Skeleton.” YouTube, www.youtube.com/watch?v=fAZDhJ3m7K5U. Accessed 27 May 2023. SARKAMARI. “Character Rigging for Beginners: 02 Skinning.” YouTube, 29 May 2020, www.youtube.com/watch?v=IEP8Fwc2YfM. Accessed 13 Dec. 2024. SARKAMARI. “Character Rigging for Beginners: 03 Primary Controls.” YouTube, 29 May 2020, www.youtube.com/watch?v=fm37fnS9MNSI. Accessed 19 Nov. 2025. SARKAMARI. “Character Rigging for Beginners: 04 Secondary Controls.” YouTube, 29 May 2020, www.youtube.com/watch?v=t8OXYOqGBNY. Accessed 7 Apr. 2026. SARKAMARI. “Character Rigging for Beginners: 05 Finalizing the Rig.” YouTube, 29 May 2020, www.youtube.com/watch?v=EDNdq9qNW7X1A. Accessed 7 Apr. 2026. SARKAMARI. “Maya Facial Rigging 06: Eyebrow Controls.” YouTube, 30 Jan. 2021, www.youtube.com/watch?v=dAAHfBCAw18. Accessed 12 Feb. 2026. Jale's Pond. “Maya 2023	How to Rig a LONG Skirt.” YouTube, 31 Jan. 2024, www.youtube.com/watch?v=wNicFiuWqD8. Accessed 19 Dec. 2025.
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Software (Closed-Source-and-Proprietary)		
Software	**Monthly ($)**	**3.5 Months**
ZBrush	$30.00	$105.00
Maya	$245.00	$857.50
Substance 3D	$59.99	$209.97
Houdini	$22.50	$78.75
Nuke (Indie est.)	$56.99	$196.00
Premiere	$22.99	$80.47
Unreal	$0.00	$0.00
Total (non-student)	$436.48	$1,527.69

In the context of professional-grade digital content creation and compositing workflows, the financial implications of utilizing industry-standard software must be carefully considered when student licensing provisions are unavailable. The applications in question—Autodesk Maya, Adobe Substance 3D, and Houdini, Foundry Nuke, Unreal Unreal Engine, Premiere —represent the apex of contemporary practice in 3D modeling, texturing, and visual effects compositing respectively.

Accordingly, the aggregate projected expenditure for non-student licensing across these three platforms amounts to **in excess of $1,500.00** for the defined period. As such, students must exercise both prudence and strategic awareness in securing access to these essential digital tools. In light of the substantial financial burden associated with full commercial licensing of Autodesk Maya, Adobe Substance 3D, and Houdini, Foundry Nuke, Unreal Unreal Engine, Premiere it is incumbent upon them to actively pursue and utilize accredited educational licensing schemes wherever possible. Furthermore, students must ensure compliance with all licensing restrictions, particularly those pertaining to non-commercial usage, while maximizing the pedagogical value derived from these platforms within the allotted timeframe. Diligent engagement, structured practice, and efficient time management become not merely advisable but essential, given that access—whether subsidized or otherwise—is both temporally limited and economically significant. In this regard, the responsible acquisition and judicious use of software resources constitute an integral component of professional development, reflecting not only technical competence but also an informed and disciplined approach to industry-standard practices.

*In addition to software subscriptions, candidates must also allocate approximately **$2,000.00 USD for essential hardware**. This includes a laptop ($1,200.00), tablet ($300.00), stereo set ($250.00), mouse ($100.00), and e-pencil ($100). These items are necessary to support the full range of course activities and media interactions.

*Lastly, **per candidate**, to enrol in this course or sit the exam, it will cost **$375.00 USD** (fee subject to change).

*$1.00USD = $160.00JMD, as of April-01-2026.

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Title-Of-Lesson-01	: Introduction
Aim	: To introduce students to the structure, scope, and workflow of the VFX course.
Objectives	: Students should be able to: 1. understand the overall course sequence and expected outputs; 2. identify the major stages of a VFX pipeline; 3. set up an organized project folder for classwork and assignments. 4. Understand the basics of photography.
Required Materials	: Computer workstation, Blender or Maya, course outline, notebook, storage device or cloud folder.
Lecture	: Overview of VFX workflows, course expectations, file naming, version control, scene organization, and the difference between modeling, rigging, animation, texturing, compositing, and look development. Introduction to shot-based production thinking.
Tutorial	: Create a basic production folder structure with subfolders for assets, scenes, textures, renders, comps, and references. Open the software interface, review workspace layouts, and save a properly named starter project.
Assignment	: Prepare a personal production workspace and submit a screenshot of the folder structure and software workspace, with a short paragraph describing the student's learning goals for the course.
Due Date	: Before Lesson 02.
Reference	: Course handbook; software interface manual; production workflow notes.

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Title-Of-Lesson-02	VFX_IntersectingCylinders_noBoolean (Loop Tools Only)
Aim	: To teach students how to construct intersecting cylindrical forms using clean topology without boolean operations.
Objectives	: Students should be able to: 1. model intersecting cylindrical geometry with loop-based methods; 2. manage origin/pivot placement and object hierarchy; 3. apply and reset transforms correctly.
Required Materials	: 3D software, modeling workspace, reference sketches, loop modeling tools.
Lecture	: Discussion of topology flow, cylindrical intersections, clean edge loops, transform application, object origins/pivots, and parenting fundamentals for controlled modeling workflows.
Tutorial	: Students create two or more intersecting cylinders, adjust origin points, parent objects where needed, and finalize geometry using loop-based editing. Emphasis is placed on maintaining clean topology and avoiding destructive shortcuts.
Assignment	: Model a small hard-surface object composed of intersecting cylinders, such as a pipe joint, valve section, or mechanical connector, with clean topology and correct transforms.
Due Date	: Before Lesson 03.
Reference	: Blender Manual: Modeling and Transformations; Autodesk Help: Pivot and Parenting Tools; class topology notes.

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Title-Of-Lesson-03	: VFX_nodes/layers_PBR_Cylinder_SprayCan
Aim	: To introduce students to layered material workflows and PBR texturing through a cylindrical spray can asset.
Objectives	: Students should be able to: 1. explain the principles of PBR materials; 2. build layered textures and material masks; 3. texture a cylindrical object with believable surface variation
Required Materials	: 3D software, shader editor, painting tools, texture references, spray can model.
Lecture	: Overview of PBR theory: base color, roughness, metallic, normals, and mask-driven layering. Discussion of decals, wear, paint breakup, label placement, and surface realism.
Tutorial	: Students texture a spray can using nodes or layer-based painting methods. The exercise includes base paint, printed label area, roughness variation, edge wear, and small imperfections for realism.
Assignment	: Create a fully textured spray can with at least three visible material layers and present clay, textured, and close-up renders.
Due Date	: Before Lesson 04.
Reference	: Blender Manual: Shader Editor and Texture Paint; Substance-style texturing notes; PBR material guide.

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Title-Of-Lesson-04	: VFX_rig&picker_Ball, BallTail, BallPendulum
Aim	: To teach the foundations of simple rigging and control pickers using ball-based objects.
Objectives	: Students should be able to: 1. create simple rigs for a bouncing ball, tail, and pendulum; 2. build basic controller relationships; 3. organize scene hierarchies for animation.
Required Materials	: 3D software, ball mesh, tail mesh, pendulum mesh, rigging tools.
Lecture	: Introduction to rigging concepts: parents and children, constraints, control objects, naming conventions, and the purpose of picker-based workflows for fast animation selection.
Tutorial	: Students rig three simple systems: a ball, a ball with an overlapping tail, and a pendulum. Controls are named, color-coded if available, and arranged for easy animation access.
Assignment	: Submit three working rigs with clearly labeled controls and a screenshot of the outliner or hierarchy.
Due Date.	: Before Lesson 05
Reference.	: Armature and constraints manual; beginner rigging notes; class rig organization guide

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Title-Of-Lesson-05	: VFX_rig&picker_Humanoid_Cylinder_Body
Aim	: To introduce basic humanoid body rigging using simple cylindrical forms.
Objectives	: Students should be able to: build a simplified humanoid skeleton; create body controls for major movement areas; test range of motion without breaking the rig.
Required Materials	: 3D software, simplified humanoid cylinder body model, rigging workspace.
Lecture	: Body rig structure, joint placement, FK/IK basics, controller design, and how simplified body rigs prepare students for animation blocking and walk cycles.
Tutorial	: Students rig a cylinder-based humanoid body with controls for pelvis, torso, arms, legs, and head. The focus is clarity, deformation testing, and animator-friendly selection.
Assignment	: Create and submit a functional body rig with demonstration poses: neutral, walk contact, and crouch.
Due Date.	: Before Lesson 06
Reference	: Character rigging manual; skeletal setup notes; control design guide.

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Title-Of-Lesson-06	: VFX_rig&picker_Humanoid_Ball_Face
Aim	: To teach simple facial rigging techniques using a ball-based humanoid head.
Objectives	: Students should be able to: 1. set up basic face controls for eyes, jaw, brows, and mouth shapes; 2. understand simple deformation approaches for facial animation; 3. organize facial controls for easy access.
Required Materials	: 3D software, simple head model, rigging tools, deformation controls.
Lecture	: Principles of facial rigging, controller placement, expression systems, eye and jaw setup, and the relationship between facial controls and performance animation.
Tutorial.	: Students build a basic face rig on a ball-head character, including jaw opening, eye aiming, eyebrow motion, and simple expression control
Assignment	: Submit a face rig showing at least five expressions: neutral, happy, sad, surprised, and angry.
Due Date	: Before Lesson 07.
Reference	: Facial rigging notes; deformation control handbook; character expression reference guide

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Title-Of-Lesson-07	: VFX_keyframes_Ball, BallTail, BallPendulum
Aim	: To introduce keyframing principles through simple animated objects.
Objectives	: Students should be able to: 1. animate a bouncing ball with convincing timing and spacing; 2. animate follow-through and overlap in a tail; 3. animate pendulum motion with arcs and natural swing behavior.
Required Materials	: Completed rigs from Lesson 04, timeline, graph editor, playback tools.
Lecture	: Keyframe fundamentals, timing, spacing, squash and stretch, overlap, follow-through, arcs, and simple graph editor cleanup.
Tutorial	: Students animate three short exercises: bouncing ball, ball with trailing tail motion, and pendulum swing. They refine key poses and improve motion curves.
Assignment	: Submit playblasts of all three animations, each showing clear motion principles and cleaned timing.
Due Date	: Before Lesson 08.
Reference	: Animation principles notes; dope sheet and graph editor manual; timing charts.

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Title-Of-Lesson-08	: VFX_keyframes_Humanoid_cycle/progressive
Aim	: To teach progressive humanoid body animation through a cycle-based workflow.
Objectives	: Students should be able to: 1. block and refine a basic humanoid cycle; 2. identify contact, down, passing, and up poses; 3. improve animation through progressive passes.
Required Materials	: Humanoid body rig, reference poses, timeline, graph editor.
Lecture	: Walk-cycle structure, body mechanics, weight shift, foot placement, arm counter-motion, and the value of progressive refinement from blocking to spline polish.
Tutorial	: Students block a simple cycle for the humanoid body, then refine posture, timing, and locomotion rhythm in stages.
Assignment	: Produce a looped humanoid cycle with readable body mechanics and submit viewport playblast plus key pose sheet.
Due Date	: Before Lesson 09.
Reference	: Walk-cycle analysis notes; body mechanics guide; animation polish checklist.

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Title-Of-Lesson-09	: VFX_keyframes_Humanoid_mixed_face&body
Aim	: To combine facial and body animation into a unified character performance.
Objectives	: Students should be able to: 1. coordinate face and body timing; 2. add expression changes to support motion; 3. create a short performance shot with clear intent.
Required Materials	: Humanoid body rig, facial rig, timeline, audio mandatory, performance notes.
Lecture	: Performance animation basics, staging emotion through pose and expression, synchronizing head movement with facial changes, and balancing clarity with simplicity.
Tutorial	: Students animate a short character moment combining body gesture and facial expression, such as noticing an object, reacting emotionally, or turning and speaking silently.
Assignment:	Create a 5–10 second performance shot integrating body movement and facial changes with a clear beginning, middle, and end.
Due Date	: Before Lesson 10.
Reference	: Performance animation notes; facial-body synchronization guide; acting for animators handout.

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Title-Of-Lesson-10	: VFX_MattePainting
Aim	: To introduce matte painting for environmental extension and shot enhancement.
Objectives	: Students should be able to: 1. explain the role of matte painting in VFX; 2. plan foreground, midground, and background separation; 3. create a simple environment extension from still imagery.
Required Materials	: Image editing software, plate image, painting tools, layered file setup.
Lecture	: Matte painting concepts, perspective matching, photo integration, atmospheric depth, color harmony, and composition for believable set extension.
Tutorial	: Students extend a plate by adding background architecture, sky, landscape, or environmental detail using layers and paint-over methods.
Assignment	: Produce a matte-painted environment extension with labeled layers and a before/after comparison image.
Due Date.	: Before Lesson 11
Reference	: Matte painting workflow notes; perspective and composition guide; digital paint handbook.

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Title-Of-Lesson-11	: VFX_CameraTrack
Aim	: To teach basic camera tracking for integrating 3D elements into live-action footage.
Objectives	: Students should be able to: 1. identify good tracking features in footage; 2. solve a basic camera track; 3. place and align 3D objects into a tracked shot.
Required Materials	: Tracking footage, 3D software with motion tracking tools, simple proxy object.
Lecture	: Principles of matchmoving, feature detection, lens considerations, error checking, solving camera movement, and grounding CG objects in live plates.
Tutorial	: Students track a short shot, clean weak markers, solve the camera, define ground orientation, and place a simple 3D object into the scene for preview integration.
Assignment	: Submit the tracked scene file, tracking screenshot, and a preview render showing one integrated CG object.
Due Date	: Before Lesson 12.
Reference	: Motion tracking manual; camera solve checklist; matchmove troubleshooting notes.

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Title-Of-Lesson-12	: VFX_Rotoscope
Aim	: To teach basic rotoscoping and mask animation for isolating moving subjects.
Objectives	: Students should be able to: 1. create and animate masks around moving elements; 2. manage edge quality and temporal consistency; 3. prepare roto mattes for compositing tasks.
Required Materials	: Footage with moving subject, compositing tools, masking tools, drawing tablet optional.
Lecture	: Rotoscoping fundamentals, shape layering, spline animation, feathering, motion consistency, and practical uses in cleanup and compositing.
Tutorial	: Students isolate a moving subject using animated masks, refine the silhouette, and output a clean matte suitable for compositing.
Assignment	: Deliver a rotoscoped matte sequence and a short composite demonstrating background replacement or element isolation.
Due Date	: Before Lesson 13.
Reference	: Rotoscoping notes; compositing mask guide; edge refinement checklist.

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Title-Of-Lesson-13	: VFX_ChromaKey
Aim	: To introduce green-screen compositing and key extraction.
Objectives	: Students should be able to: 1. explain the fundamentals of chroma key workflows; 2. extract a clean key from screen footage; 3. combine foreground and background plates with color consistency.
Required Materials	: Green-screen footage, background plate, compositing software, color correction tools.
Lecture	: Chroma key principles, spill suppression, matte cleanup, edge treatment, garbage mattes, and matching lighting and perspective between layers.
Tutorial	: Students key out a green-screen subject, suppress spill, refine the matte, and composite the subject into a replacement environment.
Assignment	:Submit a finished chroma key composite with source plate, matte preview, and final comp.
Due Date	: Compositing keying manual; spill suppression notes; plate integration guide.
Reference	: Before Lesson 14.

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Title-Of-Lesson-14	: VFX_PBR
Aim	: To deepen students' understanding of physically based rendering for believable VFX integration.
Objectives	: Students should be able to: 1. explain how PBR supports realistic asset integration; 2. build materials that respond correctly to light; 3. compare roughness, metallic, and normal detail variations.
Required Materials	: 3D software, rendered asset, HDR lighting or studio light setup, material reference boards.
Lecture	: Advanced PBR workflow, measured realism, material response, reflection behavior, lighting interaction, render consistency, and look-development testing.
Tutorial	: Students create or refine several PBR materials for a VFX asset and render the object under different lighting conditions to evaluate realism and consistency.
Assignment	: Prepare a look-development sheet showing at least PBR materials with a render and material breakdown notes.
Due Date	: Before Lesson 15.
Reference	: PBR material handbook; look-development notes; rendering and shading manual.

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Title-Of-Lesson-15	: VFX_PanoramicStitched_HDR
Aim	: To teach panoramic image stitching and HDR creation for lighting and environment capture.
Objectives	: Students should be able to: 1. explain the purpose of panoramic HDR environments in VFX; 2. stitch multiple images into a panoramic result; 3. generate or prepare an HDR environment for scene lighting.
Required Materials	: Bracketed image set, panoramic stitching software, HDR merge tools, tripod optional.
Lecture	: Panorama capture principles, exposure bracketing, stitching workflow, HDR merging, seam correction, and using panoramic maps for realistic lighting and reflections.
Tutorial	: Students organize a bracketed image sequence, stitch a panoramic environment, merge exposures where required, and test the result as a lighting source in a simple 3D scene. Assignment: Submit one stitched panoramic HDR environment, a lighting test render, and a short reflection explaining the capture and stitching workflow.
Due Date	: One week after Lesson 15 or on the final submission date set by the instructor.
Reference	: HDR imaging notes; panoramic stitching guide; environment lighting handbook.

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(PRD00) Production - "Immersive-Digital-Media-Production-IIb"

Programme	: Media Production
Department	: Media
Prerequisites	: CSEC® (Literature, Technical Drawing, VisualArts, TheatreArts, or Music) (CAPE® Performing Arts U1 and/or U2) or (satisfactory demo-reel/interview)
Type Of Course	: Production
Course Title	: "Immersive-Digital-Media-Production-IIb"
Course Code	: PRD00
Credits	: 3
Year	: Year-One (2026)
Semester	: s1-spring, s2-summer, s3-fall
Duration	: 1 Semester, 15 weeks, 60 hours (4 hours per week),(Every Wednesday@12:00am) (T1: January-01_May-01)(T2: June-01_August-31)(T3: September-01_Demember-01)
Lecturer / Tutor	: Israel Andrew Brown
Email	: israelandrewbrown@proton.me
Email (Organization)	: ioaadj@proton.me
Website	: https://github.com/israelandrewbrown/Immersive-Digital-Media-Production
Discord	: https://discord.gg/TsZG7jTQDA
Jitsi (Online-Meet)	: https://meet.proton.me/join/id-QF8T7C2RVR#pwd-oZaELJBJuukm
e-Resources	: thirty-embedded-video-links-twenty-hours-cited-throughout-the-document
Rationale	: fx
Description	: fx
Objectives	:

1. Understand fx
2. Simulate fx
3. Develop fx
4. Integrate fx

Course Assessment	: "(PRJ00) Project – Individual Project"
Equipment (Required)	: Laptop (<16gb ram) (<256gb ssd), e-Mouse, Microphone, camera, <50mbps internet, room
cost-of-course/exam	: $375.00 USD ($1.00usd to 160jmd)(fee-subject-to-change)(payable-via-bitcoin) .
.Donate Bitcoin	: bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v (request-btc-address-for-tuition)

This course material is intended for educational use and is based on the author's current understanding of the subject matter. While the author has taken reasonable steps to ensure the accuracy of the information presented, the rapidly evolving nature of artificial intelligence and media production means that some details may become outdated. The information in this book is distributed on an "As Is" basis without warranty. While every precaution has been taken in the preparation of the book, the author shall not have any liability to any person or entity with respect to any loss or damage caused or alleged to be caused directly or indirectly by the instructions contained in this book or products described in it.

Grenade-Explosion and Fountain

Progression 1
Progression 2
Progression 3

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(free-and-open-source) (7hrs)

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(PRD00) Production - “Immersive-Digital-Media-Production-llc”

Programme	: Media Production
Department	: Media
Prerequisites	: CSEC® (Literature, Technical Drawing, VisualArts, TheatreArts, or Music) (CAPE® Performing Arts U1 and/or U2) or (satisfactory demo-reel/interview)
Type Of Course	: Production
Course Title	: “Immersive-Digital-Media-Production”
Course Code	: PRD00
Credits	: 3
Year	: Year-One (2026)
Semester	: s1-spring, s2-summer, s3-fall
Duration	: 1 Semester, 15 weeks, 60 hours (4 hours per week),(Every Wednesday@12:00am) (Term: January-01_May-01)(Term: June-01_August-31)(Term: September-01_Demember-01)
Lecturer / Tutor	: Israel Andrew Brown
Email	: israelandrewbrown@proton.me
Email (Organization)	: ioaadj@proton.me
Website	: https://github.com/israelandrewbrown/Immersive-Digital-Media-Production
Discord	: https://discord.gg/TsZG7jTQDA
Jitsi (Online-Meet)	: https://meet.proton.me/join/id-QF8T7C2RVR#pwd-oZaELJBJuukm
e-Resources	: eighty-embedded-video-links-twenty-hours-cited-throughout-the-document

Rationale :

This course addresses the growing need for digital-citizens who can integrate A.I. tools into traditional media creation workflows, to prepare them for an industry/society increasingly reliant on AI-enhanced production pipelines that maintain core storytelling and technical media production skills.

Description :

This course leverages State of The Art (SoTA) free and open-source artificial intelligence models and software to produce media (text, imagery, audio, 3d-printable-mesh). Building on foundational media composition skills (storytelling, storyboarding, animatic-creation, and video-editing), students will learn methods in artificial digital asset generation. Understand and utilize digital media production techniques to effectively; create, modify & compose artificially-generated digital assets into 2d/3d-printable-media. Troubleshoot common AI-related issues in media production.

Goals :

1. Understand and utilize media production techniques to effectively generate, modify,& compose artificially generated digital assets into media.
2. Appreciate how artificial intelligence models can address and resolve inefficiencies in digital media pipelines, improving overall productivity.
3. Identify, analyze, and resolve common issues arising from the use of artificial intelligence in digital media production.
4. Recognize the limitations of automation in digital media production, recognizing which roles are more challenging to automate and why.

Equipment (Required)	: Laptop (≤16gb ram) (≤256gb ssd), e-Mouse, Microphone, camera, ≤50mbps internet, room
Equipment (Edge-A.I)	: (NVIDIA-GB10-Grace-Blackwell-Superchip) or (Apple-Silicon M-Series Chips)
Equipment (Optional)	:2xWebCam, 2xUSBC-to-C Cable, 2xTripods, 1xChArUco-Board, 1xHeadgear, Green-Screen
Course Assessment .	: Please note, knowledge gained from this course will be assessed in the “(PRJ00) Project – Individual Project” by its respective examiner.
cost-of-course/exam	: $375.00 USD ($1.00usd to 160jmd)(fee-subject-to-change)(payable-via-bitcoin)
Donate Bitcoin	: bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v (request-btc-address-for-tuition)

This course material is intended for educational use and is based on the author's current understanding of the subject matter. While the author has taken reasonable steps to ensure the accuracy of the information presented, the rapidly evolving nature of artificial intelligence and media production means that some details may become outdated. The information in this book is distributed on an “As Is” basis without warranty. While every precaution has been taken in the preparation of the book, the author shall not have any liability to any person or entity with respect to any loss or damage caused or alleged to be caused directly or indirectly by the instructions contained in this book or by the operating systems, large language models, image diffusion models, computer vision models, computer software and hardware products described in it.

Week One
(Lecture) Introduction to “Immersive-Digital-Media-Production”
What is media? What is a film? What is a “cultural-relevance” "characterization" “perspective” “pacing” ?
What is a computer? What is an operating system? What is hardware and software?
What is an algorithm? What is “Artificial Intelligence"? What is “software user documentation”?
What is a Large Language Model (LLM)? What is stable diffusion? What is cv / pose detection?
What is the “Chain-of-Thought” and “Zero-shot”? What is a ControlNet? What is a.i. Hallucination?
What is the “e-pipeline” and its roles”? What is scarcity and abundance? What is a.i-slop vs i.c.l.?
6-Cs of a.i.-asset-mining; “conscience”, “character”, “control”, “consistency”, "continuity" “confirmation” ?
What is an influence?What is “Glaze”software?What is“digital-watermarking”?What is“a.i.-asset-mining”?
What is “free-and-open source”? What are free-and-open-source software licenses?
(.txt)(.fountain)(.jpeg)(.ply)(.stl)(.usd)(.ogg)(.FLAC)(.srt)(.otio)(.glb)(.csv)(.exr)(.mkv)(.cbz)
What is “production-meeting”? What is creative “control” and “freedom”? What is “critique/approval” ?
How to inference relevant “free-and-open-source” a.i.-models? eg.“pinokio.co”“fal.ai”“replicate.com”
Assignment: Create and deploy a Studio Website or Youtube® channel.

input-to-output, i2o (definitions) (webUI[Gradio])(nodeUI[ComfyUI])(CLI)

Week Two
(Lecture) Story Development (logline, synopsis, 5-Ws, 3-Act-Structure, protonganist/antagonist)
(Lecture) Screenplay (scene, action, character, dialogue, parentheses)
(Tutorial) computer-use, browser-use, text-to-text ([deepseek-R1] ollama)

Week Three
(Lecture) Visual Development (Silhouette-Shape-Language/Line-Art/Coloured-Render)
(Tutorial) [t2i, i2i] <Characters/Props/World>([sd-v1.5] Krita-AI-Diffusion)

Week Four-Seven
(Lecture) Audio Development (Expressive-Voice-Over, Sound-Effects, Music, Subtitles)
(Tutorial) Voice_Design/Clone/Changer/csm (qwen3-VD/tts)(chatterbox-VC/tts)
(Tutorial) text-to-audio, image/video-to-audio (Stable-Audio-Open-1.0) (mmAudio)
(Tutorial) audio-to-MIDI / Split Instrumentals & Vocal (Basic-Pitch) (UVR-v5)
(Tutorial) audio-to-audio (DiffRhythm[ICL]) (ACE-Step-v1.5[ICL])
(Tutorial) upscale_audio-to-≥48kHz (lavaSR)
(Tutorial) audio-to-text (whisper-v3)

Week Eight-Nine
(Seminar) Hybrid-Composition-2d/3d (beats, lighting-diagram, pre/post-animatics, pre/post-boards)
(Critique) Record-Characterization (reference, audition, performance, dramatization, takes, critique)

Week Ten
(Lecture) Mesh Development (Retopology/Remesh)
(Tutorial) Image-to-3dMesh/PBR <Characters/Props/ >(Tripo-SR/SG/SF)
(Tutorial) Image-to-3dMesh/PBR <Characters/Props/ >(Trellis-v2)
(Tutorial) Image-to-3dMesh/PBR < / /World>(HunyuanWorld-v2.0)

Week Eleven
(Lecture) Look Development (UV/TextureTransfer-via-ReBake) (3-Point-Lighting)
(Tutorial) 3dMesh-to-PBR/NPR <Characters/Props/World>([sd-v1.5]StableGen)(hunyuan3D-2.1)

Week Twelve
(Lecture) Armature Development (pivots, child/parent, controllers, orient, bind, skin, weight, test)
(Lecture) What is a “Non-Deform/Deform-Rig” ?
(Tutorial) Auto-Rig (‘rig-gns’ plugin for Blender-v5.0) ([Rigify-HumanMetaRig]Template)

Week Thirteen
(Lecture) Motion Development (detect/calibrate-cameras, record, process-data, view-data)
(Tutorial) “Face” - Motion Capture ([MediaPipe] Deadface-v1.0)
(Tutorial) “Body” - Motion Capture ([MediaPipe] FreeMoCap-v2.0)
(Tutorial) Retarget Rig_FMC/Rig_Faceit to Rig_Rigify
(Tutorial) layer “face”+“body” motion-capture-performance

Week Fourteen
(Lecture) Motion Development (VFX) (FX) ([cuda/ROCm/mlx_required][cloud])
(Tutorial) Image-composite/edit [t2i, i2i, ii2i] ([Z-Image,Z-Image-Edit])
(Tutorial) Video, Video-Reference [t2v, i2v, i2vWa, a2v, v2v](LTX-2)
(Tutorial) 3D-Camera-Estimation ([Colmap,Golmap] camera-tracker-v4)
(Tutorial) ChromaKey/Segmentation ([sam-hq]RotoForge-AI)(CorriderKey)
(Tutorial) Re-Light (map[*albedo,normal,depth,metallic,roughness*]) ([MVInverse][DepthAnything-v2])(ctdr)
(Tutorial) sdr-to-hdr, text/in/out_Painting-to-paramaraic-i/v (CubeComposer)(DiT360)(ltx-hdr-LoRa)
(Tutorial) upscale_image/video-to-≥4k (NextGenUp [real-ESRGAN])

Week Fifteen :
(Critique) review, view or showcase of completed projects.

Course Review : https://github.com/israelandrewbrown
Donate Bitcoin : bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v

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Title-Of-Lesson-02 : **Story Development**

Aim : To introduce the fundamental principles and structural elements required for developing a coherent and engaging story.

Objectives :

1. Define and write a compelling one-sentence logline.
2. Summarize a story idea into a concise synopsis.
3. Apply the 5-Ws (Who, What, Where, When, Why) to build a story's foundation.
4. Explain and outline a narrative using the 3-Act-Structure.
5. Create and describe the core traits of a protagonist and antagonist.
6. Understand the basic formatting of a screenplay: scenes, action, and dialogue.

Required Material : computer, internet access, access to ollama (`deepseek-R1`), Trelby.

Lecture :

1. Story Development: Key concepts including logline, synopsis, the 5-Ws, the 3-Act-Structure, protagonist, and antagonist.
2. Screenplay: An introduction to screenplay formatting, covering scene, action, character, dialogue, and parentheses.

Tutorial :Browser-use and text-to-text generation: Using Ollama (with the `deepseek-R1` model) to assist in the creative writing process.

Assignment : Develop an original story concept and present its core components. Your submission must include:

1. A one-sentence logline.
2. A one-paragraph synopsis (approx. 150-250 words).
3. A short description of your protagonist and antagonist.
4. A brief outline based on the 3-Act-Structure.
5. A 3-page screenplay based on treatment.

Due Date : Week 3

References:

Khan Academy. “Using the 5 Ws and the Bonus H to Make Summaries | Reading | Khan Academy.” *YouTube*, 20 Feb. 2025, www.youtube.com/watch?v=T3LsopJ4dec. Accessed 26 Feb. 2025.

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StudioBinder. “The Anatomy of a Screenplay Part 2 Formatting Techniques to Elevate Your Script.” *YouTube*, 18 Dec. 2023, www.youtube.com/watch?v=KUnclLt2RqQ.

StudioBinder. “How to Write a Script for TV — Anatomy of a Screenplay Part 3.” *YouTube*, 22 Apr. 2024, www.youtube.com/watch?v=h4imidyLecA.

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Title-Of-Lesson-03 : **Visual Development**

Aim : To introduce the visual design pipeline, from initial abstract shapes to a fully rendered concept, using both traditional techniques and AI-powered tools.

Objectives :

1. Understand/apply Silhouette design to create strong, recognizable identities.
2. Utilize Shape Language (e.g., circles, squares, triangles) to communicate personality and function of a character, prop, or environment.
3. Refine a chosen silhouette into detailed Line-Art.
4. Produce a Coloured Render to communicate final materials, lighting, mood.
5. Use text-to-image (t2i) and image-to-image (i2i) tools to generate and iterate on visual concepts.

Required Materials: computer, software:Krita, access to Stable-Diffusion (`sd-v1.5`)

Lecture :

1. Silhouette: Starting with basic shapes to test for recognizability/visual interest.
2. Shape Language: Using fundamental shapes to build and define the design.
3. Line Art: Using line art to refine and clarify the design.

Tutorial : Visual Concept [t2i, i2i]: A session using the Stable-Diffusion (`sd-v1.5`).

1. Students will practice using text-to-image (t2i) to generate initial ideas for Characters, Props, or Worlds.
2. Students will explore image-to-image (i2i) to refine their own rough sketches or previously generated images.

Assignment : Students will derive sheets: bios, turnarounds, action-poses, face-expressions, from their finished characters, world and props, title card design using the above mentioned techniques.

Due Date : Week 4

Reference:

Tyler Edlin. “Mastering the Design Pipeline.” *YouTube*, 2024, https://www.youtube.com/watch?v=_15SIWLGQEU&ab_channel=TylerEdlin.

Tyler Edlin. “The BEST Environment Design EXERCISE for BEGINNERS.” *YouTube*, 2019, https://www.youtube.com/watch?v=mhvtuZIEV_Q&ab_channel=TylerEdlin.

Tyler Edlin. “DESIGN BETTER CHARACTERS : Essential Fundamentals.” *YouTube*, 2024, https://www.youtube.com/watch?v=9AgGCtfbuLs&ab_channel=TylerEdlin.

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Title-Of-Lesson-04-07 : **Audio Development**
(Week_04-07)

Aim : To develop a comprehensive understanding of the audio post-production pipeline and gain practical skills in generating, manipulating, and integrating all elements of a soundscape using a suite of modern AI-powered tools

Objectives :

1. Articulate the narrative function of voice-overs, sound effects, music, subtitles.
2. Generate high-quality, expressive dialogue using text-to-speech (TTS) and modify vocal performances with voice changers (VC) and explore conversational speech models (csm).
3. Create original SFX from text prompts using text-to-audio models.
4. Analyze existing audio by converting it to MIDI (audio-to-MIDI) for remixing.
5. Deconstruct songs into separate instrumental and vocal stems for audio mixing.
6. Apply audio-to-audio style transfer techniques to alter the rhythm/instrumentation of a track.
7. Automatically/accurately transcribe dialogue to create subtitles (audio-to-text).

Required Materials: Microphone, computer, software: Audacity, Aegisub.

Lecture :

1. Expressive Voice-Over:
   Analyzing tone, pacing, emotion. The difference between narration and character dialogue.

2. Sound Effects:
   Designing a soundscape.

3. Music:
   The psychology of music in the media. Scoring, themes, and mixing music with dialogue.

4. Subtitles:
   Legal requirements and creative best practices for captioning and subtitles.

Tutorial :

1. Expressive Voice-Over
   e-Tools : `qwen3-VD/tts`, `chatterbox-VC/tts`,
   Activities : Generating character dialogue using text-to-speech. Using voice-changer technology to alter pitch, timbre, and emotion or pre-existing recordings.
   Technique: Altering compression

2. Sound Effects
   e-Tools : `Stable-Audio-Open-1.0`, `mmAudio`
   Activities : Using text-to-audio to generate specific sound effects.
   Technique: Altering reverb.

3. Music
   e-Tools : (`DiffRhythm`[ICL], `ACE-Step-v1.5`[ICL])
   Activities: Experimenting with audio-to-audio models to change the style of a beat or melody.
   Technique:Running a melody through audio-to-MIDI (Basic-Pitch) to extract the musical notes for use in a different instrument. Using UVR-v5 to split a song into its instrumental and vocal tracks.

4. Subtitles & Transcription
   e-Tools : `whisper-v3`, `lavaSR`
   Activities: Using Whisper-v3 to automatically transcribe all dialogue from a video clip and format it into a subtitle file.
   Technique: Editing a subtitle file.

Assignment : Create the audio-design for the screenplay created from earlier assignment.

Due Date : Week 8

StudioBinder. “The Science of Sound in Film — Film Sound Recording for Beginners.” *YouTube*, 29 July 2024, www.youtube.com/watch?v=mXtnHHJFREM.

Goobster's Room. “Every Major Audio Effect Explained in 8 Minutes!” *YouTube*, 16 Mar. 2025, www.youtube.com/watch?v=JGGnla8RKP4. Accessed 12 Apr. 2025.

Insider Tech. “How to Make Your Voice Sound Better in Audacity.” *YouTube*, 14 Aug. 2021, www.youtube.com/watch?v=P7u5Aw9VadM. Accessed 10 Dec. 2025.

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Title-Of-Lesson-08 : **Hybrid-Composition-2d/3d**

Aim : To master the process of non-destructively-editing and translating a written story into a hybrid dynamic visual sequence, planning camera work, composition, timing using 2d/3d visuals assets.

Objectives :

1. Identify/illustrate key narrative Story Beats and lighting diagrams as colored panels.
2. Create non-destructively, a detailed Shot List that describes the visual plan for a scene.
3. Create non-destructively, a clear Storyboard (.ppt-software), using visual conventions for character blocking, direction of movement, and camera work.
4. Assemble storyboard panels into a timed Animatic with placeholder audio and camera movement (pan, tilt, zoom), cuts (transitions) to guide the viewer's eye.
5. Use 2d/3d assets/space to create non-destructively, a hybrid-animatic.

Required Materials: computer, e-tablet, e-pencil, stereo-set, e-mouse, e-tool: Krita, Storyboarder, Blender-v5.0.0.

Lecture :

1. Lighting-Diagram/Beats: Identifying the most critical emotional moments or plot points and visualizing them as Story-Beats (key images).
2. Pre-Boards vs. Post-Boards: Moving from rough thumbnail sketches (pre-boards) to cleaner, more detailed panels from animatics (post-boards). Understanding shot types (e.g., Wide, Medium, Close-up) and framing rules to create visually appealing and clear images.
3. Pre-Animatics vs. Post-Animatics: Moving from motion -panels to hybrid-2d-3d-animatics.
4. Camera Movement: Planning 2D and 3D camera moves (pan, tilt, zoom, dolly, truck) to add dynamism and perspective in 3d space.
5. Cuts & Transitions: Understanding the pacing and emotional impact of different edits (e.g., hard cut, L-cut, J-cut, cross-dissolve).

Tutorial :

Activity 1: Storyboarding:
Students will draw the Pre-Storyboard panels, paying attention to colour-coding characters and using arrows to indicate direction of movement and camera movement.

Activity 2: Animatic Assembly:
Students will create scenes based on their panels; non-destructively in 3d space. Timing each panel / keyframe, add digital 2d/3d camera movement in 3d/2d space), and sync the visuals with the sound effects and expressive voice-over (from Weeks 4-7).

Activity 3: Shot List:
Students will take their synopsis (from Week 2) and create a detailed Shot List for a short scene, including scene number, shot type, description, and planned camera movement.

Assignment :

1. Lighting-Diagram/Story Beats: 1 colored panel of the most important moments of your scene.
2. Shot List: (Scene No., Shot No., Shot Type, Description, CM, Location).
3. Pre-Storyboard and Post-Animitics: A multi-panel storyboard for the chosen scene.
4. Pre-Animatic and Post-Animatic: A 1-15 minute video file of your panels with audio (VO, SFX, Music).

Due Date : Week 9

StudioBinder. “Ultimate Guide to Camera Shots: Every Shot Size Explained.” *YouTube*, 4 May 2020, www.youtube.com/watch?v=AyML8xuKfoc.

StudioBinder. “Directing Camera Movement Filmmaking Techniques for Directors: .” *YouTube*, 30 July 2018, www.youtube.com/watch?v=GbnYBmqBbKA.

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WerwackFx. “Werwack/Storyliner-Support” *GitHub*, 12 June 2025, www.github.com/werwack/storyliner-support.

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Spitfire Storyboards. “‘ubisoft shot manager’ (now ‘storyliner’) (previs, hybrid + storyboard tools).” *YouTube,* 16 Oct. 2022, https://www.youtube.com/watch?v=8yBjm6ojwe8.

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Title-Of-Lesson-09: **Characterization**

Aim : To explore the principles and techniques for developing and recording a believable and compelling character performance, moving from static concept to dynamic execution.

Objectives :

1. Analyze professional performances using reference material.
2. Understand dramatization as the art of making a character's internal thoughts and conflict externally visible.
3. Explain the iterative production concept of recording multiple takes to achieve the best performance.
4. Learn how to give and receive constructive critique on performance choices.

Required Materials: ≤1 camera, computer

Lecture :

This lecture transitions from character *design* to character *performance*, focusing on the techniques used in acting and motion to bring a character to life.

Audition:
A trial hearing given to a singer, actor, or other performer to test suitability for a project, employment, professional training or competition

Reference:
The "actor's research." How to study real-world people, animals, and existing performances to inform a character's unique movement, tics, and expressions.

Performance:
This is the combination of all choices (voice, body language, timing, expression) that create a believable personality. We will analyze how small choices create a big impact.

Dramatization:
How to visually or audibly express subtext. If a character is lying, how do you *show* it (e.g., avoiding eye contact, fidgeting) instead of just saying it?

Takes:
The professional workflow of iterating on a performance. This applies to voice acting (recording multiple line readings) and animation (creating different versions of a motion).

Critique:
The art of the feedback loop. How to articulate *why* a performance is or isn't working using specific, actionable language rather than vague notes.

Assignment : Create and record a facial and body performance for each character based on your animatics.

Due Date : Week 10

Pierrick Picaut. “Master Production Workflow in Blender!” *YouTube,* 18 Feb. 2026, www.youtube.com/watch?v=JckDEblB7gc. Accessed 20 Feb. 2026.

Pierrick Picaut. “Everything about Linking & Overrides in Blender.” *YouTube*, 27 Jan. 2026, www.youtube.com/watch?v=z-KH-zmV79I. Accessed 20 Feb. 2026.

Pierrick Picaut. “Shot, Scene, Sequence Explained Simply.” *YouTube*, 12 Jan. 2026, www.youtube.com/watch?v=lHqKbB6_Nqc. Accessed 20 Feb. 2026.

ExplainingComputers. “Blender Video Editing: Learn Workflow, Transitions, Modifiers, More!” *YouTube*, 14 Sept. 2025, www.youtube.com/watch?v=99ZdOuJMrao. Accessed 26 Feb. 2026.

Title-Of-Lesson-10: **Mesh Development**

Aim : To introduce the fundamentals concept of 3D-mesh-topology/retopology and explore AI-driven methods for generating 3D models from 2D images.

Objectives :

1. Define retopology/explain its importance for animation and game performance.
2. Differentiate between:
   a high-poly (sculpted/scanned) mesh,
   a low-poly (animation/game-ready) mesh.
3. Use image-to-3DMesh generator to generate models.
4. Create basic 3D assets for Characters, Props, or Worlds based on 2D-art.
5. Analyze the topology of an AI-generated mesh and identify its limitations.
6. 3D-printing applications.
7. Setup 3d-workspace for retopology.

Required Materials: computer, e-mouse, access to 3d-model-generator..
2D concept art (e.g., from Week 3) to be used as input images.
3D model viewing software (e.g., Blender).

Lecture :
Topology, the arrangement of vertices, edges, and faces that form a mesh/structure.
Retopology, the process of recreating a "clean," efficient, low-polygon mesh over the top of a "messy," high-polygon mesh (which often comes from 3D sculpting or scanning.

Tutorial :
e-Tools: 256^3, 512^3, 1024^3, 1536^3; 3d-mesh generator.
Activity: This is a practical workshop on generating 3D assets from 2D images.

1. Using concept art (Characters, Props, or Worlds) and using it as input for the AI models.
2. We'll compare the outputs from each tool, discussing speed, model quality, mesh density.
3. Perform remesh/retopology on a mesh.

Assignment :
Generated Models: Share screenshots of the 3D models generated from at one of the specified AI tools (eg. `Tripo-SR`, `Tripo-SG`, `Tripo-SF` or `Trellis-v2.0`).
Include at least one screenshot showing the 3D model's wireframe.

Due Date : Week 11

Digitalist. "Every 3D Concept Explained (Basics)." *YouTube*, 15 Sept. 2025, www.youtube.com/watch?v=p3MU7J-0284. Accessed 26 Nov. 2025.

Digitalist. "3D Modelling Explained (Basics)." *YouTube*, 6 Oct. 2025, www.youtube.com/watch?v=bChytx8PHvU.

Institute for Manufacturing (IfM), University of Cambridge. "What Is 3D Printing?" *YouTube*, 21 Dec. 2017, www.youtube.com/watch?v=bcTzyx35odY.

Blender Bob. "Intersecting Cylinders in Blender." YouTube, 24 Aug. 2023, www.youtube.com/watch?v=79d-QFMxQm8. Accessed 17 Mar. 2026.

Stefan 3D AI Lab. "Ep.1 Mesh Editing (Sculpting) - from AI to Game-Ready." *YouTube*, 9 Oct. 2025, www.youtube.com/watch?v=ouHL_Rlebss. Accessed 26 Nov. 2025.

Stefan 3D AI Lab. "Ep.2 Mesh Optimization (Retopology) - from AI to Game-Ready." *YouTube*, 16 Oct. 2025, www.youtube.com/watch?v=-5zAPGdYjts. Accessed 26 Nov. 2025.

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The Joy of Blender. "Blender Character Tutorial - Part 6 (Mouth & Teeth)." *YouTube*, 22 Jan. 2024, www.youtube.com/watch?v=R4GPER0C5As.

Ryan King Art. "How to Use the Boolean Modifier in Blender (Tutorial)." *YouTube*, 4 June 2024, www.youtube.com/watch?v=CHqH5oz0DvQ.

Ryan King Art. "Understanding Object Origins in Blender." *YouTube*, 2 Oct. 2023, www.youtube.com/watch?v=9gn_1V1sCS8. Accessed 9 Dec. 2025.

Ryan King Art. "Understanding Parenting in Blender." *YouTube*, 22 Dec. 2023, www.youtube.com/watch?v=5oclP4hQndc. Accessed 22 Nov. 2024.

FredericCervini. "Maya to Blender Tips: How to Move Your Pivot Point (Origin Point) like in Maya." YouTube, 15 June 2022, www.youtube.com/watch?v=7hCHlaf7t5k. Accessed 20 Feb. 2026.

My CG Tutor. "How to Freeze Transforms in Blender (like Maya)." *YouTube*, 19 Dec. 2024, www.youtube.com/watch?v=jo_myEEIDt0. Accessed 15 Dec. 2025.

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Title-Of-Lesson-11: **Look Development**

Aim : To introduce the "Look Development" pipeline, covering the technical preparation of 3D model-textures (UVs, baking, edit) and the application of modern AI tools to generate final textures.

Objectives :

1. Explain the critical role of UV Unwrapping in the texturing process.
2. Differentiate between PBR (Physically Based Rendering) and, NPR (Non-Photorealistic Rendering).
3. Utilize AI-driven models (e.g., `sd`, `hunyuan3D-2.1`, `StableGen`) to generate PBR or NPR texture maps for 3D meshes.
4. Apply generated textures to various asset types, including characters, props, and world elements via Texture-Edit.
5. Examine a toon-shader.
6. Describe the workflow for Texture-Transfer/Re-Baking (e.g., high-poly detail to low-poly map).
7. Create 3-Point-Lighting to the assets/textures.

Required Materials: computer, e-tablet, e-pencil, e-tool: Blender-v5.0.0

Lecture :

UV Unwrapping principles, the process of Texture-Transfer from one model to another, and "Re-Baking" essential maps (like normal, ambient occlusion, or curvature). Texture-Edit in external e-painting application (eg. Krita, Photoshop).

Tutorial :

1. Applying generative models (`sd`, `hunyuan3D-2.1`, `StableGen`) to create textures, for <Characters/Props/World> assets in both PBR and NPR styles.
3. Transfer textures between 3d-meshes.

Assignment :

Students must take a provided 3D model (e.g., a prop or a simple character) and use an AI tool (`sd`, `hunyuan3D-2.1`, `StableGen`) to generate a complete PBR texture set. They will then apply/transfer these textures to the model and submit a final Render.

Due Date : Week 12

Digitalist. “3D Materials Explained (Basics).” *YouTube*, 19 Oct. 2025, www.youtube.com/watch?v=abo6ikU-Lx8. Accessed 26 Nov. 2025.

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Stefan 3D AI Lab. “Ep.3 UV Map & Normal Baking - from AI to Game-Ready.” *YouTube*, 21 Oct. 2025, www.youtube.com/watch?v=wCZcYKI-tGg. Accessed 26 Nov. 2025.

Stefan 3D AI Lab. “Ep.4 Texturing (Painting) - from AI to Game-Ready.” *YouTube*, 28 Oct. 2025, www.youtube.com/watch?v=L_uPx5z4y5c. Accessed 26 Nov. 2025.

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Title-Of-Lesson-12 : **Armature Development**

Aim : To understand and build the technical rig required to bring a static 3D model to .. life, distinguishing between rigs for hard-surface objects and organic characters. Create a rig/shapekeys with automated tools for retargeting.

Objectives :

1. Define rigging and explain its role in the 3D animation pipeline.
2. Exam a simple non-deform/deform rig for a mechanical/organic object by setting . . pivots and parent/child, armature hierarchies.
3. Exam controllers to manipulate a rig.
4. Explain the concept of skinning/binding and weight painting for organic models.
5. Use an auto-rig template (like Rigify) to generate a "meta-rig" for a character.
6. Use an auto-rig-face template (like Faceit) to generate "52-shapekeys" for character.
7. Test a rig by posing it to find and fix deformation errors.

Required Materials: computer, e-mouse, e-tool: Blender-v5.0.0

Lecture : .

This theory-focused lecture covers the manual process and concepts of building . a rig from scratch.Setting pivots, establishing child/parent hierarchies, creating . . controllers, joint orientation, the binding/skinning process, painting skin weights, . and testing the rig for good deformation. A rig is the "digital skeleton" and control . system that allows an animator to pose and move a 3D model.

. . 1. Non-Deform (Mechanical) Rigs: For objects that don't bend. Relies on pivots and hierarchies.
. 2. Deform (Organic) Rigs: For objects that bend. Relies on a skeleton, skinning, and weighting.

Tutorial :

Using a template (specifically: Rigify-HumanMetaRig, Faceit) to quickly generate a . . complex, production-ready rig with 52-Artkit-blendshapes (shape-keys) for a bipedal . . character.

Assignment :

Using the Rigify-HumanMetaRig template, students will rig a provided bipedal . . 3D model. They must then pose the character in three distinct poses and submit .. screenshots to demonstrate the rig is functional.

Due Date : Week 13

Carlino, Joey. “Rigging for Impatient People - Blender Tutorial.” *Youtube*, 5 May 2023, www.youtube.com/watch?v=DDeB4tDVCGY.

Matthew DeSa Art. “Rigging Ball and Tail in Blender Livestream.” *YouTube*, 12 Mar. 2024, www.youtube.com/watch?v=nNEx36i5jI8. Accessed 29 Nov. 2025.

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Richstubbsanimation. “Amazingly EASY Way to Rig Characters in Blender 4.0.” *Youtube*, www.youtube.com/watch?v=jIwrswJEFBQ.

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Blender Studio. “Blender Studio Rigging Tools - Blender Studio.” Blender Studio, 2023, studio.blender.org/training/blender-studio-rigging-tools/. Accessed 16 Mar. 2026.

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Eve Sculpts.“Blender Tutorial - Use Rigify to Easily Rig Your Characters (Part 1).” *Youtube*, Dec 10, 2021 www.youtube.com/watch?v=umtig82NGdk.

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Title-Of-Lesson-13 : **Motion Development**

Aim : To capture realistic human performances using accessible motion capture tools and apply .. that data to the custom character rigs created in the previous week.
Objectives :
1. Understand the end-to-end motion capture pipeline: detect, calibrate, record, process, and view data.
2. Use Deadface-v1.0 to capture facial expressions and head movement.
3. Use FreeMoCap-v2.0 to capture full-body motion.
4. Define and execute retargeting to transfer mocap data from a source skeleton to their Rigify rig.
5. Layer and combine separate face and body animation data into one unified performance.

Required Materials : 1xcamera, Laptop (≤16gb ram) (≤256gb ssd), e-Mouse, Microphone, ≤50mbps internet, room
Optional Materials : 2xWebCam, 2xUSB-C Cable, 2xTripods, 1xChArUco-Board, 1xHeargear

Lecture :
Calibrate Cameras:
The process of setting up the capture environment so the software understands the 3D space.

Record Footage:
Capturing the actor's performance from one or more angles.

Process Data:
The "solving" phase, where the software analyzes the video footage and generates 3D coordinates or a solver skeleton.

View Data:
Cleaning and reviewing the captured data for any noise, jitter, or errors before applying it to the final character.

Tutorial 1: “Face” - Motion Capture (e-Tool: Deadface-v1.0)
- Activity: Students will use their webcams to capture their own facial performance. This will involve recording a few lines of dialogue or a series of strong expressions and exporting the resulting animation data.

Tutorial 2: “Body” - Motion Capture (e-Tool: FreeMoCap-v2.0)
- Activity: Students will set up a camera and record themselves performing a simple, clear full-body action . They will then process this footage through FreeMoCap to generate the 3D body motion data.

Tutorial 3: Retargeting (Face & Body) (Faceit x BlendarMoCap)
- Activity: This is the core technical step of applying the captured data.
  - Body: Students will import their FreeMoCap data and use a retargeting interface to map the motion from the mocap skeleton onto their character's Rigify rig.
  - Face: Students will connect the data from FaceLandmarkLink to the facial controls on their Rigify rig.

Tutorial 4: Layering the Performance
- e-Tool: 3D Software's animation editor (e.g., Blender's Non-Linear Animation Editor) (or Add-on)
- Activity: Students will combine the two separate animation files.
- Process: The body motion will be set as the base animation layer. The facial animation will then be added as a separate layer on top, allowing both to play simultaneously and create one complete performance.

Assignment : using the techniques discussed in class, create two mcp-s from your animatic.
Due Date : Week 14

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Title-Of-Lesson-14 : **VFX Integration Development**

Aim : To explore advanced motion development techniques (VFX/FX) and introduce a comprehensive suite of AI-driven tools for generative media, 3D-motion-composition enhancement.

Objectives : Create video.

Required Materials:
- A computer with a dedicated GPU or Online-Demo-Space
- Stable internet access (for [online-demo-space])
- Access to the specified AI models and software
- (`Z-Image`, `LTX-2`, `sam`, `ColmapxGolmap`, `CorriderKey`, `MVInverse/c-t-d-r`, `DiT360`, `ESRGAN`)

Lecture : A deeper dive into Visual Effects (VFX) and Special Effects (FX)

Tutorial :

1. Image Generation (t2i, i2i):
2. Video Synthesis (t2v, i2v, v2v, a2v, i2vWa):
3. 3D-Camera-Estimation:
4. ChomaKey/Segmentation
5. AI Re-Lighting: Manipulating PBR maps (albedo, normal, depth, metallic, roughness) with tools.
6. Panoramic/HDRI Generation: Using text/inPainting/outPainting for 360° environments.
7. AI Upscaling: Enhancing image/video resolution to >4K.

Assignment : Create a video using vfx-replacement/influence-techniques to alter characters, props, … background, sim-effects, lighting,

Due Date : Week 15

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Category	Company	Model Name	Links (Demonstration Spaces)
text	*Deepseek-AI*	deepseek-R1	https://chat.deepseek.com
Imagery	*Stability-AI*	Stable-Diffusion	https://github.com/Acly/krita-ai-diffusion
audio Expressive Voiceover	*Alibaba* *Resemble-AI*	Qwen3-VD/tts chatterbox-VC/tts	https://huggingface.co/spaces/Qwen/Qwen3-TTS https://huggingface.co/spaces/ResembleAI/Chatterbox
Sound Effects	*Stability-AI* *Sony*	Stable-Audio-Open-v1.0 MMAudio	<DemontrationSpace> https://huggingface.co/spaces/hkchengrex/MMAudio
Music	*ACE-Studio* <*company*> *Spotify*	ACE-Step-v1.5[ICL] UVR-v5 Basic-Pitch	https://huggingface.co/spaces/ACE-Step/Ace-Step-v1.5 https://github.com/Anjok07/ultimatevocalremovergui https://basicpitch.spotify.com/
Upscaler	<*company*>	lavaSR	https://huggingface.co/spaces/YatharthS/LavaSR
Subtitles	*open-AI*	Whisper-v3	https://huggingface.co/spaces/openai/whisper
Mesh	*VAST-AI-Research*	Tripo-SR <Tier-01 (256^3)> Tripo-SG <Tier-02 (512^3)> Tripo-SF <Tier-03 (1024^3)>	https://huggingface.co/spaces/stabilityai/TripoSR https://huggingface.co/spaces/VAST-AI/TripoSG <DemontrationSpace>
	Microsoft *Tencent*	Trellis-v2 <wildcard (1536^3)> HunyuanWorld-2.0	https://huggingface.co/spaces/microsoft/TRELLIS.2 <DemontrationSpace>
Rig	<*company*>	Rigify (Rig-GNS)	https://22722studios.gumroad.com/l/avmsyt
Movement	*FreeMoCap*	FreeMoCap Deadface	https://github.com/freemocap/freemocap https://github.com/Qaanaaq/DeadFace
imagery video chromakey re-lighting panorama	*Alibaba* *Lightricks* *CorridorCrew* *NVIDIA* *Insta360*	Z-Image LTX-2 CorridorKey-v1.0 MVInverse / c-t-d-r DiT360	https://chat.qwen.ai/ https://ltx.studio/ <DemontrationSpace> https://huggingface.co/spaces/maddog241/mvinverse-demo https://huggingface.co/spaces/Insta360-Research/DiT360

Free-And-Open-Source Generative Artificial Intelligence Models (Software User Documentation)		
Model Name	**License**	**Official Github Repository**
deepseek-R1	MIT-License	https://github.com/deepseek-ai/DeepSeek-R1
Stable-Diffusion-1.5	MIT-License	https://github.com/Acly/krita-ai-diffusion
Qwen3-VD/tts chatterbox-VC/tts	Apache-2.0-License MIT-License	https://github.com/QwenLM/Qwen3-TTS https://github.com/resemble-ai/chatterbox
Stable-Audio-Open-1.0 MMAudio	Stability-AI-Community-License MIT-License	https://github.com/Stability-AI/stable-audio-tools https://github.com/hkchengrex/MMAudio
ACE-Step-v1.5[ICL] UVR-v5 Basic-Pitch	MIT-License MIT-License Apache-2.0-License	https://github.com/ace-step/ACE-Step-1.5 https://github.com/Anjok07/ultimatevocalremovergui https://github.com/spotify/basic-pitch
lavaSR	Apache-2.0-License	https://github.com/ysharma3501/LavaSR
whisper-v3	MIT-License	https://github.com/openai/whisper
Tripo-SR <Tier-01 (256^3)> Tripo-SG <Tier-02 (512^3)> Tripo-SF <Tier-03 (1024^3)> Trellis-v2.0 <(1536^3)> HunyuanWorld-2.0	MIT-License MIT-License Tencent-Community-License	https://github.com/VAST-AI-Research/TripoSR https://github.com/VAST-AI-Research/TripoSG https://github.com/VAST-AI-Research/TripoSF https://github.com/microsoft/TRELLIS.2 https://github.com/Tencent-Hunyuan/HY-World-2.0
(Mediapipe)(MMPose)(ocv) [FreeMoCap]	Apache-2.0-License	https://github.com/google-ai-edge/mediapipe https://github.com/open-mmlab/mmpose https://github.com/opencv/opencv
Rigify	GPL-3.0-License	https://github.com/cessen/rigify
Z-Image, LTX-Video-2 Corridorkey MVInverse / c-t-d-r DiT360 diffHDR	Apache-2.0-License LTX2-Community-License Custom-License NC / Apache-2.0-License MIT-License	https://github.com/Tongyi-MAI/Z-Image https://github.com/Lightricks/LTX-2 [t2v, i2v, i2vWa, a2v, v2v] https://github.com/nikopueringer/CorridorKey https://github.com/nv-tlabs/cosmos1-diffusion-renderer https://github.com/Insta360-Research-Team/DiT360 https://github.com/gbymat/DiffHDR-pytorch

code-agent	*OpenCode*	MIT License	https://github.com/anomalyco/opencode/

*Registered candidates for the course *"Immersive-Digital-Media-Production"* are allowed to inference, either *online* or *locally*, the *free and open-source* a.i. models, of which permit *commercial and research-use*.

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Proprietary - Artificial Intelligence Online Tools (November 2025)			
Category	**Company / Organization**	**Services**	**Official-Website-Link**
text	*Anthropic AI*	-text-generation	https://www.anthropic.com/
Imagery (4k)	*Midjourney AI*	-text-to-image -image-to-image -image-edit	https://www.midjourney.com/
Audio (48 kHz)	*Elevenlabs AI* *Suno AI*	-Expressive VoiceOver (VC/tts/csm) -Sound Effects -Music -Subtitles	https://elevenlabs.io/ https://suno.com/home
mesh (4096^3)	*VAST AI* *World Labs AI*	-Mesh-Generation -Auto-Remesh -Texture-Generation (4k) -Mesh-to-Texture -Mesh-Segmentation -Auto-Rigging	https://www.tripo3d.ai/ https://app.modddif.com/ https://www.worldlabs.ai/
movement	*Rekoko (Vision)*	-Facial-Motion-Capture -Body-Motion-Capture	https://www.rokoko.com/
video (4k)	*Runway AI* *LumaLab AI* *Higgsfield AI*	-text-to-video-Generation -video-to-video-Generation -Character-Replacement	https://runwayml.com/ https://lumalabs.ai/ https://higgsfield.ai/
re-lighting (4k)	*Beeble AI*	-Green-Screen-Removal -PBR-Map-Estimation -SDR-to-HDR	https://www.switchlight.beeble.ai/
code-agent	*Openai*	agentic-coding	https://openai.com/codex/

*Candidates registered for the course or the *"exam-only"* are allowed to inference either the *free and open-source* or *closed-source* a.i. models.

*Monthly subscription fees for each service provider range from $25.00 USD to $100.00 USD, on average. These subscriptions grant comprehensive access to all required functionalities and media categories—**text, imagery, audio, mesh, movement, video, relighting**—necessary to complete the course, associated projects, or the exam-only track.

*For planning purposes, candidates should note that the instructional period spans 15 weeks (approximately 3.5 months) and requires access to services from 10 ten distinct providers. Accordingly, individuals pursuing the course or the *"exam-only"* pathway are advised to allocate a total budget (**for 3.5 months**) of **no less than $875.00 USD** and up to $3,500.00 USD, depending on provider selection and subscription tier, to ensure continuous access to the proprietary, closed-source AI tools referenced above.

*In addition to software subscriptions, candidates must also allocate approximately **$2,000.00 USD for essential hardware**. This includes a laptop ($1,200.00), tablet ($300.00), stereo set ($250.00), mouse ($100.00), and e-pencil ($100). These items are necessary to support the full range of course activities and media interactions.

*Lastly, **per candidate**, to enrol in this course or sit the exam, it will cost **$375.00 USD** (fee subject to change).

*$1.00USD = $160.00JMD, as of December-13-2025.

*Blender3D. https://www.blender.org/

Donate Bitcoin : bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v

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(PRD00) Production - “Immersive-Digital-Media-Production-III”

Programme	: Media Production
Department	: Media
Prerequisites	: CSEC® (Literature, Technical Drawing, VisualArts, TheatreArts, or Music) (CAPE® Performing Arts U1 and/or U2) or (satisfactory demo-reel/interview)
Type Of Course	: Production
Course Title	: “Immersive-Digital-Media-Production-III”
Course Code	: PRD00
Credits	: 3
Year	: Year-One (2026)
Semester	: s1-spring, s2-summer, s3-fall
Duration	: 1 Semester, 15 weeks, 60 hours (4 hours per week),(Every Wednesday@12:00am) (T1: January-01_May-01)(T2: June-01_August-31)(T3: September-01_Demember-01)
Lecturer / Tutor	: Israel Andrew Brown
Email	: israelandrewbrown@proton.me
Email (Organization)	: ioaadj@proton.me
Website	: https://github.com/israelandrewbrown/Immersive-Digital-Media-Production
Discord	: https://discord.gg/TsZG7jTQDA
Jitsi (Online-Meet)	: https://meet.proton.me/join/id-QF8T7C2RVR#pwd-oZaELJBJuukm
e-Resources	: thirty-embedded-video-links-twenty-hours-cited-throughout-the-document

Rationale :

The demand for rigorous, pipeline-driven post-production workflows is paramount. This course bridges the gap between raw creative output and final, engine-ready deployment. By mastering open-source and industry-standard interchange formats (`.gltf`, `.aaf`, `.otio`, `OpenColorIO`, `git`), students will build robust technical foundations. This curriculum is designed to equip undergraduates to share production-ready files to MSc-level professionals and environments , ensuring they can engineer flawless post-production pipelines for game engines, film, and interactive web environments.

Description :

This intensive 15-week course explores the advanced technical processes of post-production. Students will migrate assets into game engines utilizing the `.gltf` format, mastering meshes, textures, rigs, and animations. The course then transitions into audio conforming and mastering using `.aaf` workflows for voiceover, sound effects, and music. Finally, students will explore editorial metadata via `.otio`, color science using `.ocio` (ACES 2.0), and deploy text-based websites to document their pipelines using `Git` version control.

Objectives :

1. Master the `.gltf` workflow for exporting optimized 3D meshes, textures, rigs, and animations reliably into standard game engines.
2. Execute professional audio mastering techniques within an `.aff` finishing workflow, specifically balancing voice-over, sound effects, and music tracks.
3. Implement advanced video editing and color grading pipelines utilizing `.otio` for editorial handoff and `.opencolourio` (ACES 2.0) for color-managed finishing.
4. Utilize `.git` version control to initialize, develop, commit, and deploy a text-only portfolio website, demonstrating readiness for MSc-level progression or independent industry mastery.
5. Synthesize these discrete technical skills into a unified, professional-grade post-production pipeline.

Course Assessment : “(PRJ00) Project – Individual Project”

Equipment (Required) : Laptop (<16gb ram) (<256gb ssd), e-Mouse, Microphone, camera, <50mbps internet, room

cost-of-course/exam : $375.00 USD ($1.00usd to 160jmd)(fee-subject-to-change)(payable-via-bitcoin) .
.Donate Bitcoin : bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v (request-btc-address-for-tuition)

This course material is intended for educational use and is based on the author's current understanding of the subject matter. While the author has taken reasonable steps to ensure the accuracy of the information presented, the rapidly evolving nature of artificial intelligence and media production means that some details may become outdated. The information in this book is distributed on an “As Is” basis without warranty. While every precaution has been taken in the preparation of the book, the author shall not have any liability to any person or entity with respect to any loss or damage caused or alleged to be caused directly or indirectly by the instructions contained in this book or products described in it.

IoAaDJ_3d-Generalist__ByWeek_LiveDemontration

1. Introduction
2. .gltf_mesh
3. .gltf_textures
4. .gltf_rig
5. .gltf_aniamtion
6. .aff_vo
7. .aff_sfx
8. .aff_music
9. .otio
10. .opencolourio
11. git_initiate
12. git_develop
13. git_commit
14. git_deploy
15. Review

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Title-Of-Lesson-01	Week_01_Introduction
Aim	To introduce the complete post-production pipeline covering open standards and handoff workflows required for immersive digital media.
Objectives	Define the roles of .gltf, .aff, .otio, and .opencolourio in modern post-production. Understand the requirements for MSc-level progression versus undergraduate understanding. Set up local directory structures and install required open-source software.
Required Materials	Computer workstation, internet access, notebook, external storage drive.
Lecture	Overview of post-production life cycles. The importance of interoperability and open standards to prevent vendor lock-in. Introduction to the term handoff.
Tutorial	Software installation and workspace directory configuration.
Assignment	Create a pipeline setup document outlining your hardware specifications and configured software paths.
DueDate	End of Week 01
Reference	Reference: https://www.youtube.com/watch?v=KTPdNUGwIGc

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Title-Of-Lesson-02	: Week2_.gltf_mesh
Aim	: To master the export and optimization of 3D geometry for real-time engines.
Objectives	: Optimize polygon counts, generate LODs (Levels of Detail), and export clean .gltf meshes.
Required Materials	: 3D DCC (Blender/Maya), Game Engine (Unity/Unreal), sample high-poly models. Lecture: Polygon budgets and real-time rendering constraints.
Tutorial	: Using a Retopologized high-poly mesh and exporting via .gltf.
Assignment	: Export three varying LODs of a single asset into a game engine without errors.
DueDate	: Week 3
Reference	: Khronos Group .gltf documentation;

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Title-Of-Lesson-03	Week_03_.gltf_textures
Aim	To bake and export physically based rendering (PBR) materials alongside 3D meshes.
Objectives	Bake procedural and high-resolution textures down to optimized image maps. Configure a PBR material node tree compatible with the .gltf specification. Export a fully textured model and verify material integrity in the engine.
Required Materials	3D creation software, image editing software.
Lecture	PBR workflows (Albedo, Roughness, Metallic, Normal). Channel packing for memory optimization in game engines.
Tutorial	Baking textures in Blender and exporting the compiled .gltf with embedded or external textures.
Assignment	Submit a textured asset with baked PBR maps correctly displaying in an engine viewport.
Due Date	End of Week 03

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Title-Of-Lesson-04:	Week_04_.gltf_rig
Aim:	To format and export skeletal hierarchies and skin weights suitable for real-time engines.
Objectives:	1. Structure a skeletal rig that conforms to game engine constraints (e.g., root bones, hierarchy limits). 2. Clean and normalize vertex weight painting.
Required Materials:	Export a rigged, non-animated mesh using .gltf. 3D creation software, character mesh.
Lecture:	Differences between deform bones, control bones and vertex weighting in real-time environments.
Tutorial:	Prepping a complex rigged model for export and debugging bone errors upon import.
Assignment:	Submit a rigged character model loaded in a game engine, demonstrating correct bone deformation and zero hierarchy warnings..
Due Date:	End of Week 04

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Title-Of-Lesson-05	: Week5.gltf_animation
Aim	: To integrate dynamic motion and blend shapes into the real-time environment.
Objectives	: Bake non-linear animation clips, manage NLA tracks, and export morph targets/blend shapes.
Required Materials	: 3D DCC, rigged and animated character model.
Lecture	: Animation data compression and interpolation across software. Tutorial: Baking keyframes and setting up an animation controller in-engine.
Assignment	: Trigger an idle-to-walk animation cycle and a facial blend shape inside the game engine.
Due Date	: Week 6

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Title-Of-Lesson-06-08:	Week6_.aff_vo/sxf/music (Note: .aaf refers to Advanced Authoring Format)
Aim	: To manage dialogue stems and metadata transfer from an NLE to a DAW.
Objectives	: Export timeline data via .aaf, relink source media, and perform basic vocal clean-up.
Required Materials	: NLE (Premiere/Resolve), DAW (Pro Tools/Reaper), raw dialogue stems.
Lecture	: The anatomy of the .aaf format and industry-standard dialogue mixing levels (LUFS).
Tutorial	: Exporting a sequence from an NLE, importing into a DAW, and applying a noise gate/EQ.
Assignment	: Conform a 1-minute dialogue sequence in a DAW via .aaf and normalize to -23 LUFS.
Due Date	: Week 7

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Title-Of-Lesson-09:	Week_09_.otio
Aim:	To utilize .otio (OpenTimelineIO) for seamless editorial timeline exchange and video editing workflows.
Objectives:	Understand the XML/JSON structure behind OpenTimelineIO. Export an editorial timeline from one non-linear editor (NLE) and import it into another. Relink offline media seamlessly via the .otio structure.
Required Materials:	Video editing software supporting OTIO (e.g., DaVinci Resolve, Kdenlive, Premiere), raw video clips.
Lecture:	The history of EDL, XML, and AAF, leading to the Academy Software Foundation's .otio standard for editorial handoffs. Tutorial: Cutting a short sequence, exporting the .otio file, and successfully rebuilding the cut in a secondary software environment.
Assignment:	Submit an .otio file alongside a low-res proxy render demonstrating a successful round-trip workflow.
Due Date:	End of Week 09

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Title-Of-Lesson-10:	Week_10_.opencolourio
Aim:	To implement cinematic color grading using .opencolourio and the ACES 2.0 color management framework.
Objectives:	1. Set up an ACES 2.0 color-managed workflow using OpenColorIO. Understand Input Device Transforms (IDT) and Output Device Transforms (ODT). 2. Perform primary and secondary color grading in a scene-referred linear space.
Required Materials:	Color grading software (e.g., DaVinci Resolve, Blender compositing), log-profile video footage.
Lecture:	Color science, the limitations of display-referred workflows (sRGB/Rec.709), and the advantages of the Academy Color Encoding System (ACES).
Tutorial:	Configuring .opencolourio profiles, mapping a log camera file to ACEScg, grading, and outputting to Rec.709.
Assignment:	Submit a short video featuring a distinct color grade, accompanied by node/layer screenshots of your ACES 2.0 pipeline.
Due Date:	End of Week 10

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Title-Of-Lesson-11:	Week_11_.git_initiate
Aim:	To initialize a version-controlled environment using .git for a text-only portfolio website.
Objectives:	Install Git and configure global user variables. Initialize a local repository and understand the staging area. Link the local repository to a remote platform (e.g., GitHub).
Required Materials:	Command Line Interface (Terminal/Git Bash), Code Editor (e.g., VS Code), GitHub account.
Lecture:	Introduction to version control, distributed workflows, and the importance of portfolio archiving.
Tutorial:	Running git init, setting up .gitignore, and connecting to a remote repository origin.
Assignment:	Provide a screenshot of your terminal showing a successfully initialized and linked remote .git repository.
Due Date:	End of Week 11

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Title-Of-Lesson-12:	Week_12_.git_develop
Aim:	To author the HTML structure for a .git-for-text-only-websites portfolio showcasing post-production assets.
Objectives:	1. Draft semantic HTML5 to structure portfolio sections (About, 3D Assets, Audio, Video). 2. Embed lightweight descriptions and links to cloud-hosted post-production deliverables. 3. Maintain clean, valid markup ready for version tracking.
Required Materials:	Code Editor, local development browser.
Lecture:	Semantic web design, accessibility, and designing a fast, text-only academic/professional presence.
Tutorial:	Writing the index.html file, structuring data, and verifying local browser rendering.
Assignment:	Complete the draft of your text-only HTML portfolio encompassing all semester projects.
Due Date:	End of Week 12

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Title-Of-Lesson-13:	Week_13.git_commit
Aim:	To track development history through proper .git staging, committing, and branching.
Objectives:	1. Add file changes to the git staging area. 2. Write clear, semantic commit messages detailing HTML developments. 3. Push local commits to the remote repository.
Required Materials:	Command Line Interface, local HTML repository.
Lecture:	The Git lifecycle (Working Directory, Staging, Local Repo, Remote Repo). Best practices for conventional commit messaging.
Tutorial:	Executing git add, git commit -m, and git push. Resolving minor syntax edits with new commits.
Assignment:	Generate a repository history containing at least 5 meaningful, chronological commits outlining site development.
DueDate:	End of Week 13
Reference:	https://www.youtube.com/watch?v=RgVClxjPgHM

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Title-Of-Lesson-14:	Week_14.git_deploy
Aim:	To deploy the text-only portfolio website live using static hosting via GitHub Pages.
Objectives:	Configure repository settings to enable static page deployment. Understand the Continuous Integration/Continuous Deployment (CI/CD) pipeline for static sites. Verify public access and functionality of the live web portfolio.
Required Materials:	GitHub account, finalized .git repository.
Lecture:	Web hosting principles, DNS, and static vs. dynamic web serving using GitHub Pages.
Tutorial:	Activating GitHub Pages from the main branch and forcing a deployment build.
Assignment:	Submit the live URL of your deployed text-only post-production portfolio.
Due Date:	End of Week 14

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Title-Of-Lesson-15:	Week_15Review
Aim:	To consolidate all post-production competencies and finalize the digital portfolio for assessment and future academic/professional use.
Objectives:	Review the end-to-end pipeline covering .gltf, .aff, .otio, and .opencolourio. Perform quality assurance (QA) on the deployed .git portfolio. Outline personal pathways for MSc-level continuation or independent mastery.
Required Materials:	Fully deployed portfolio, all semester project files.
Lecture:	Future-proofing digital skills, staying updated with Khronos Group and ASWF standards, and transitioning to advanced study (MSc).
Tutorial:	Peer review of live portfolios. Debugging broken links and assessing audio/video/3D deliverable quality.
Assignment:	Final submission of the complete Post-Production Digital Portfolio package.
Due Date:	End of Week 15
Reference:	Review of all previous course references.

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(PRJ00) Project – Individual/Group Project

Individual/Group Short Film (1minutes - 15minutes) (250 marks)
Pre-Production, Production and Post-production, Marketing and Distribution.
(By Group [≥6 memebers], or Individually produce a short film. This Project has five [5] sections.)
(Yearly-Due-Date: June-01/August-01/December-01) (Yearly-Result-Date:June-30/August-31/December-31)
(To document the project create five [5] folders, for the sections of the project)
(share Cloud-Drive permissions to either of the email addresses.)
Email: <ioaadj@proton.me> <israelandrewbrown@proton.me> <israelandrewbrown@gmail.com>

Section One - Pre-production (20%)
Construct a film-preproduction-package (.pdf or .cbz) containing the following:

1. Funding (Greenlight Statement - Proof-Of-Funding)
2. Expenditure (Gantt-Chart, Human resources, Software/Equipment and Budget)
3. Research-Paper/Presentation (When, Where, Who, Why, What)
4. Treatment (Author, Title, Log-line, Synopsis, Characters)
5. Screenplay (three-act-structure)(use standard-screenplay-software)(3pages ≤ *x* ≤ 15pages)
6. Concept Art (Title-Card)
   Concept Art (characters - two [≤2]) (antagonist and protagonist)
   - 1 x bio,
   - Turnarounds x 5-views, or; t-pose/a-pose views; or; front and back,
   - 6 x face expressions,
   - 1x or; 6x action poses,

   Concept Art (props)
   Concept Art (environment [hemisphere]) (set [cube]) (segmentation) (composition)
7. Story-Beats/Lighting-Diagram (coloured-images of important events/panels of scene)
8. Pre/Post-Animatic(panel/keyframe, camera movement, sound effects and expressive voiceover)
9. Pre/Post-Storyboard (colour coded characters, direct of movement, camera movement)
10. Shot List (scene no., shot no.,shot type, shot descr., camera movement, Location)
    [Presentation Of “Pre-production”.]

Section Two - Production (20%)
Catalogue (.zip)and(.txt) of the file & file-structure for the following artificially generated digital assets:

1. Asset_Text - (Text-Concept) (.txt)(.fountain)
2. Asset_Imagery - (Visual-Concept) (.jpeg)(.exr)(.mkv)
3. Asset_Audio - (Expressive-Voiceover, Sound Effects, Music, Subtitles) (.FLAC)(.srt) (.aaf)
4. Asset_Mesh - (Character-Prop-World [Mesh x PBR-Texture]) (.usd)(.stl)(.glb)(.ply)
5. Asset_MotionCaptureData - ([2d-pose/3d-pose] [Input data, output data]) (.csv)(.mkv)
   [Presentation of the file & file-structure for the all digital assets catalogure.] (.otio)

Section Three - Post-production (20%)
Create a; Physically-Based-Rendering (PBR) or Non-Photorealistic-Render (NPR) styled film (.pdf/.cbz)**:**

1. Modification of “artificially generated digital assets”.
2. Composition of “artificially generated digital assets” into a complete shot.
3. Utilization of FX, VFX (part-scene-replacement, relighting), to enhance film composition.
4. Arrangement of shots through video editing and color grading.
5. Compilation of shots as a singular sequence.
   [Presentation of the all finalized rendered story-beat/s as an image/images.]

Selection Four - Marketing (20%)
Create and deploy a website or Youtube® Channel (.pdf / .cbz) for the studio.

1. Brand Identity (logo, font, colour-palette, etc.)
2. Poster (YouTube Thumbnail) (Cover Art) (Title Card)
3. Studio Information (Home, About, Contact [Studio and Workers], Blog, Merch [optional])
4. Credits Sequence (video and spreadsheet)
5. Research (Abstract, Methodology, Results, Discussion, References)
   <Grant, Direct-Promotion, Indirect-Promotion, Rebate, Tax-Credit.>
   [Presentation of the webpage.]

Section Five - Distribution (20%)
[Optional Distribution through Film Festival]
YouTube® Channel Creation - Upload in order. (H.264/H.265 codec) (≥720p). (5 x .mkv)

1. Teaser - Title Card (Thirty [10] seconds)
2. Teaser (Ten-Thirty [10-30] seconds)
3. Trailer (one [1] minute - three [3] minutes)
4. Behind-the-scenes Documentary (15-60 minutes)
   a. Teaser at the beginning and title card at end
   b. Footage of some of the work being done.
   c. Processes and visual breakdown of artificially generated digital assets.
   d. Interview of the project manager/director/lecturer.
5. Completed Short Film (1-15 minutes)
   [There should be five (5) videos on the channel/profile playlist at completion.]

Donate Bitcoin: bc1qdjsljjzj4x83v28ks0l3cvwdkqvhyfggzc2w8v

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Section One - Pre-production (20%)		Exam-Only [ ]
Component	**Criteria-Checklist**	**Score**
[ ] Research	[ ] What, [ ] Why, [ ] Who, [ ] Where, [ ] When [ ] Sources Cited [ ] Visuals Used [ ] Gammar [ ] Mood-Board (visuals-used-on-one-page) [ ] Professionalism	/10
[ ] Treatment	[ ] Title/Author [ ] Log-line Summary - [ ] Synopsis: Act 1, [ ] Synopsis: Act 2, [ ] Synopsis: Act 3, [ ] Protagonist Bio-Summary [ ] Antagonist Bio-Summary [ ] Grammar [ ] Correct Format [ ] Maximum-Two-Pages	/10
[ ] Screenplay	[ ] Dedicated Software Used [ ] Scene Headings, [ ] Action Lines, [ ] Dialogue [ ] Official-Screenplay-Format [ ] Act 1 Setup, [ ] Act 2 Conflict, [ ] Act 3 Resolution [ ] Relevant-to-treatment [ ] Gammar	/10
[ ] Character Concept	[ ] Protagonist Bio [ ] Antagonist Bio Protagonist - [ ]Turnaround, [ ] Face Expressions, [ ] Action Poses, Antagonist - [ ]Turnaround, [ ] Face Expressions, [ ] Action Poses, [ ] Technical-Consistency [ ] Annotation	/10
[ ] Prop Environment	[ ] InteriorSet-Design-Sheet-Segmentation, [ ] InteriorSet-Deign-Sheet-Cube, [ ] ExteriorSet-Design-Sheet-Segmentation, [ ] ExteriorSet-Design-Sheet-Hemisphere, [ ] Variation, [ ] Functionality, [ ] Painted/Toon/PBR, [ ] Relevance, [ ] Format, [ ] Colour, [ ] Annotation	/10
[ ] Shot-List	[ ] Scene/Shot No., [ ] Shot Type, [ ] Shot Description, [ ] Camera Movement [ ] File-Format, [ ] Completeness, [ ] Clarity/Organization, [ ] Variety [ ] Physiological-Narrative-Function, [ ] Physical-Feasibility	/10
[ ] StoryBoard	[ ] Visual Clarity [ ] Matches Script, [ ] Storytelling, [ ] Historically-Correct-Pacing [ ] Continuity, [ ] Composition [ ] Color-Coding, [ ] Character Movement Arrows, [ ] Camera Movement Arrows, [ ] Format	/10
[ ] Storybeats Animatic	[ ] Includes Story-Beats [ ] Story-board (Indictaction-Of-Direction) Temporary - [ ] Voiceover, [ ] Sound FX, [ ] Music, [ ] Subtitles, [ ] Audio-Sync [ ] Movement-Of-Panels (Animation) [ ] Use of 2d/3d assets/space [ ] Technical Quality	/10
[ ]Expenditure	[ ] Gantt Chart (Schedule) [ ] Human Resources (Roles), [ ] Software, [ ] Equipment List [ ] Budget, [ ] Cash-Flow-Projection, [ ] Profit-And-Loss-Statement, [ ] Error-Free [ ] Format	/10
[ ] Funding	[ ] Singular-Copyright-Holder-Document, (OptionA - 10 marks) [ ] Receipts/Agreement (OptionB - 10 marks)	/10
Total		/100

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Section Two - Production (20%)			Exam-Only [ ]
Component	**Summarized-Criteria**	**Detailed-Criteria-Checklist**	**Score**
Text	Organized text assets in specified formats (.txt, .fountain).	[ ] Prompt Clarity & Effectiveness [ ] Proper Syntax (.txt /.foun / .pdf) [ ] Adheres to Pre-Production Docs [ ] Low-Hallucination [ ] Grammar	/5
Imagery (4k)	High-res visual assets matching screenplay (.jpeg, .mkv).	[ ] Resolution: Minimum resolution:720x400px/ Maximum resolution:1920×1080 (HD). [ ] Coherent w/ Concept Art [ ] Stylized [ ] Colored-Renders [ ] Variety of Shots/Angles	/5
Audio (48kHz)	Lossless audio (.FLAC) and timed subtitles (.srt).	[ ] Expressive Voiceover [ ] Subtitle Sync & Accuracy (.srt) [ ] Sound Design (SFX) [ ] Music [ ] Audio: 48kHz sound PCM (only uncompressed) MP3 AAC (if possible > 160 kBit/s)	/5
Mesh	3D models with clean topology, UVs, and PBR textures.	[ ] Optimized-Mesh-Retopology [ ] UV-Unwrapping, High-Quality PBR Textures [ ] Altered-Transforms [ ] Pivot, Manipulators [ ] Parent/Child, Hierarchy	/5
Motion-Capture-Data	Complete mocap sets: input video (.mkv) and output data (.csv)(.mkv).	[ ] Input Video (.mkv) [ ] Output Data (.csv) (.mkv) [ ] facial and body, capture layered [ ] Matches Storyboard/Animatic [ ] Usable Data (Applies to Rig)	/5
Total			/25

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Section Three - Post-production (20%)			Exam-Only [ ]
Component	**Criteria Summary**	**Detailed-Criteria-Checklist**	**Score**
Modification	Manually cleaned, refined, and documented assets to be production ready.	[ ] Fix Grammatical Errors [ ] Mesh Cleanup & Retopology [ ] Texture Adjustments & Painting [ ] Rig Refinements [ ] Mocap Data Cleaning & Smoothing [ ] Audio Mastering & Mixing [ ] Visual Asset Enhancement [ ] Seamless Integration of Changes [ ] Technical Problem-Solving [ ] Consistency with Art Direction	/10
Composition	Complete film edit matching storyboard/storybeats and script.	[ ] Logical Scene Assembly [ ] Effective Shot Sequencing [ ] Adheres to Storyboard [ ] Flow Screenplay Narrative [ ] Consistent Pacing & Rhythm [ ] Effective Camera Work in Scene [ ] Seamless Asset Integration [ ] Maintains Continuity [ ] Establishes Emotional/Atmosphere/Mood [ ] Coloured-Render	/10
Integration (FX, VFX)	Technically integrated and synchronized visual effects.	[ ] VFX (Re-lighting/Lighting) [ ] VFX (Object/Prop Replacement) [ ] VFX (Character Replacement) [ ] VFX (Background Replacement) [ ] VFX (3D-Camera-Tracking) Use any of the five listed FXs: [1 mark each] - [ ] Frost, - [ ] Fog, - [ ] Rain, - [ ] Lighting, - [ ] Smoke, - [ ] Fire, - [ ] Dust, - [ ] Gravity/Anti-Gravity	/10
Compilation (Video-Editing)	Fully edited, color-corrected, and graded film.	[ ] Effective Pacing & Timing [ ] Purposeful Transitions [ ] Use-Of-Cuts [ ] Audio/Video Sync [ ] Audio Music-Mix [ ] Consistent Color Palette [ ] Color Enhances Mood [ ] Correct Exposure & Contrast [ ] Use of Color [ ] Non-Noisey Final Cut	/10
Presentation	Final film exported to technical specs with credits.	[ ] Uploaded in standard formats such as .mp4, .mkv, .mov, or .mpeg. [ ] Codec: MJPEG H264 [ ] Resolution: Min:720x400px/ Max:1920×1080 (HD). [ ] File Size: The film file should generally not exceed 8 GB. [ ] Audio/Video Code: Multiplexed files (audio, video, subtitles in file) . [ ] Progressive (deinterlaced) footage is recommended. [ ] Audio: 48kHz sound PCM (only uncompressed) MP3 AAC (> 60 kBit/s) [ ] Subtitles: The subtitles have to be integrated (burned in) in the image. [ ] Duration: Entries must be at least 60 seconds ≥ x ≥ 15 minutes.. [ ] No watermarks or social handle text on screen)	/10
Total			/50

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Section Four - Marketing (20%)			Exam-Only [ ]
Component	**Criteria Summary**	**Criteria-Checklist**	**Score**
Brand Identity	Brand guide PDF with vector logo, colors, and fonts.	[ ] Logo Design [ ] Logotype/Font [ ] Color Palette [ ] Color and Non-Color Render [ ] .pdf	/5
Promotional Artwork	Poster, thumbnail, and title card created to spec.	[ ] Poster Composition & Impact [ ] YouTube Thumbnail (Clear, Clickable) [ ] Title Card Design [ ] High-Resolution-Poster (above 720p) [ ] Correct Aspect Ratios (2:3, 16:9, 4:3)	/5
Studio Website	Functional, responsive website with all required pages.	[ ] Home Page: [ ] About Page: Contains at least 100 words about the studio. [ ] Contact Page: Includes a functional mailto: link or contact form. [ ] Blog Page: Contains at least one post. [ ] Navigation: A navigation bar is present on all pages.	/5
Credits Sequence	Video credits sequence accompanied with spreadsheet.	[ ] Video: Credits are included in the final film video file. [ ] Music: Credit roll has a music track.t. [ ] All team members are included in the spreadsheet. [ ] Formatting: Spreadsheet has columns for "Role"& "Name". [ ] Logos: Required partner/software logos are present.	/5
Promotion & Financial Strategy	Strategy document with research, plans, and budget.	5-mark awarded for research/documentation of one the following: - Grant Research (Suitable options) - Direct-Promotion Research (e.g., ads) - Indirect-Promotion Research - Rebate Research - Tax-Credit Research [ ] Abstract, [ ] Methodology, [ ] Results, [ ] Discussion, [ ] Reference	/5
Total			/25

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Section Four - Distribution (20%)			Exam-Only [ ]
Category	**Summary**	**Criteria-Checklist**	**Marks**
Title Card Teaser (10 seconds)	Branded YouTube channel with 10s title card teaser uploaded.	[ ] Files must be uploaded in standard formats such as .mp4, .mkv, .mov, or .mpeg. [ ] Codec: MJPEG H264. [ ] Resolution: Minimum resolution:720x400px/ Maximum resolution:1920×1080 (HD). [ ] File Size: The film file should generally not exceed 8 GB. [ ] Audio/Video Code: Multiplexed files (audio, video, subtitles in a single file) preferred. [ ] Progressive (deinterlaced) footage is recommended. [ ] Audio: 48kHz sound PCM (only uncompressed) MP3 AAC (if possible > 160 kBit/s) [ ] Subtitles: The subtitles have to be integrated (burned in) into the image. [ ] Duration: Entries must be at least 60 seconds ≥ x ≥ 15 minutes.. [ ] No watermarks or social handle text on screen [ ] Deduction of category marks awarded, if video uploaded is not relevant to treatment. Video URL: www.youtube.com/	/10
Teaser (10-30 seconds)	10-30s non-spoiler video teaser uploaded.	[ ] Files must be uploaded in standard formats such as .mp4, .mkv, .mov, or .mpeg. [ ] Codec: MJPEG H264. [ ] Resolution: Minimum resolution:720x400px/ Maximum resolution:1920×1080 (HD). [ ] File Size: The film file should generally not exceed 8 GB. [ ] Audio/Video Code: Multiplexed files (audio, video, subtitles in a single file) preferred. [ ] Progressive (deinterlaced) footage is recommended. [ ] Audio: 48kHz sound PCM (only uncompressed) MP3 AAC (if possible > 160 kBit/s) [ ] Subtitles: The subtitles have to be integrated (burned in) into the image. [ ] Duration: Entries must be at least 60 seconds ≥ x ≥ 15 minutes. [ ] No watermarks or social handle text on screen [ ] Deduction of category marks awarded, if video uploaded is not relevant to treatment. Video URL: www.youtube.com/	/10
Trailer (1-3 minutes)	1-3 minute story-based trailer uploaded.	[ ] Files must be uploaded in standard formats such as .mp4, .mkv, .mov, or .mpeg. [ ] Codec: MJPEG H264. [ ] Resolution: Minimum resolution:720x400px/ Maximum resolution:1920×1080 (HD). [ ] File Size: The film file should generally not exceed 8 GB. [ ] Audio/Video Code: Multiplexed files (audio, video, subtitles in a single file) preferred. [ ] Progressive (deinterlaced) footage is recommended. [ ] Audio: 48kHz sound PCM (only uncompressed) MP3 AAC (if possible > 160 kBit/s) [ ] Subtitles: The subtitles have to be integrated (burned in) into the image. [ ] Duration: Entries must be at least 60 seconds ≥ x ≥ 15 minutes. [ ] No watermarks or social handle text on screen [ ] Deduction of category marks awarded, if video uploaded is not relevant to treatment. Video URL: www.youtube.com/	/10
Behind-the-Scenes Doc. (<15 minutes)	BTS video (<7m) with all required segments uploaded.	[ ] Uploaded: The BTS is the fourth video on the channel. [ ] Content Req. 1: Includes footage of work being done. [ ] Content Req. 2: Includes visual breakdown of five types of assets. [ ] Content Req. 3: Includes an interview segment. [ ] Structure: Includes an opening title card. [ ] Length: The video runtime is over 15 minutes. [ ] Audio: Interview audio is clear and audible. [ ] SEO: Optimized with unique title, description, and tags. [ ] Quality: Uploaded in at least 720p. [ ] Thumbnail: A custom thumbnail is uploaded. [ ] Deduction of category marks awarded, if video uploaded is not relevant to treatment. Video URL: www.youtube.com/	/10
Completed Short Film (1-15 minutes)	Final film (1-15m) with captions and end screen uploaded.	[ ] Files must be uploaded in standard formats such as .mp4, .mkv, .mov, or .mpeg. [ ] Codec: MJPEG H264. [ ] Resolution: Minimum resolution:720x400px/ Maximum resolution:1920×1080 (HD). [ ] File Size: The film file should generally not exceed 8 GB. [ ] Audio/Video Code: Multiplexed files (audio, video, subtitles in a single file) preferred. [ ] Progressive (deinterlaced) footage is recommended. [ ] Audio: 48kHz sound PCM (only uncompressed) MP3 AAC (if possible > 160 kBit/s) [ ] Subtitles: The subtitles have to be integrated (burned in) into the image. [ ] Duration: Entries must be at least 60 seconds ≥ x ≥ 15 minutes. [ ] No watermarks or social handle text on screen [ ] Deduction of category marks awarded, if video uploaded is not relevant to treatment. Video URL: www.youtube.com/	/10
Total			/50

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Exam-Only [ ]

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Section	Marks (Awarded)	Marks (Maxium)	Percentage Score = (Awarded ÷ Max) × 100	Weight (20%)	Final Grade = (Percentage × 0.20)
1 - Pre-production		100		20%	
2 - Production		25		20%	
3 - Post-Production		50		20%	
4 - Marketing		25		20%	
5 - Distribution		50		20%	
Totals		250		100%	Final Grade

Please note, to use this:

1. Plug in the marks awarded.
2. Divide by the “maximum marks” to get the percentage.
3. Multiply by 0.20 (since each section = 20%).
4. Add all five contributions to get the final grade (out of 100).

GRADING SCHEME

Grade	% Equivalent	Level of Pass
A	90 – 100	Honours
A-	80 – 89	Honours
B+	75 – 79	Credits
B	70 – 74	Credits
B-	66– 69	Credits
C+	60 – 65	Pass
C	55 – 59	Pass
C-	50 – 54	Acceptable Pass
D	45 – 49	Fail - Supplemental
F	44 – 0	Fail

Candidate:	TRN: 000-000-000	Name:	Signature:	DD/MM/Year
Candidate:	TRN: 000-000-000	Name:	Signature:	DD/MM/Year
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Plugins *(Compatible with "Blender-v5.0.0")*

StoryLiner (Animatics) *(GPL-3.0-License)* https://superhivemarket.com/products/storyliner	**collection-opacity-adjuster** *(GPL-3.0-License)* https://github.com/israelandrewbrown/collection-opacity-adjuster
AutoCam *(GPL-3.0-License)* https://renderrides.gumroad.com/l/autocam	**Camera-Shakify** *(GPL-3.0-License)* https://github.com/ra100/camera_shakify

Sculpt-Layers (CC-BY-4.0) https://superhivemarket.com/products/sculpt-layers	**UV-Flow** (UV-Unwrapper) *(GPL-2.0-License)* https://superhivemarket.com/products/uv-flow
QRemeshify (Quad-Remesher) *(GPL-3.0-License)* https://github.com/ksami/QRemeshify	**Retopoflow** (Retopology) *(GPL-3.0-License)* https://github.com/CGCookie/retopoflow

Ucupaint (Textures) *(GPL-3.0-License)* https://github.com/ucupumar/ucupaint	**Quick Edit** (Blender-Krita_Bridge) *(GPL-3.0-License)* [natively-built-into-Blender]

CorridorKey *(GPL-3.0-License)* https://github.com/nikopueringer/CorridorKey	**3D-Camera-Tracker** (≤-v4) *(GPL-3.0-License)* https://superhivemarket.com/products/camera-tracker
RotoForge-AI *(GPL-3.0-License)* https://github.com/MagnumVD/RotoForge-AI	**HDRI-cube-Converter** *(GPL-3.0-License)* https://github.com/PlasteredCrab/BlenderCubemapConverter
Switch.Light.Beeble.AI https://beeble.ai/resources/downloads	**HDRI-dome-Projector** *(GPL-3.0-License)* https://superhivemarket.com/products/hdri-dome-projection
3D-Gaussian-Splatting https://github.com/TimChen1383/EasyEnv	**zForm/Depth Map Batch** (image-to-depthmap) *(GPL-3.0-License)* https://superhivemarket.com/products/depth-map-batch-for-images

Rigify *(GPL-3.0-License)* [natively-built-into-Blender]	**RigGNS** (Mixamo-Alternative) (Rigify) *(GPL-3.0-License)* https://22722studios.gumroad.com/l/avmsyt
Easy Pivot *(GPL-3.0-License)* https://superhivemarket.com/products/easypivot	**Transport-Rig** *(GPL-3.0-License)* https://blenderigmaster.gumroad.com/l/vrf_plus
Easy IK *(GPL-3.0_License)* https://superhivemarket.com/products/quick-ik	**Easy Skinning** *(GPL-3.0-License)* https://superhivemarket.com/products/voxel-heat-diffuse-skinning
Easy Weights *(GPL-3.0_License)* https://extensions.blender.org/add-ons/easyweight/	**Easy Bone-Widget** *(GPL-3.0-License)* https://github.com/waylow/boneWidget
Easy Set-Driven-Key *(GPL-3.0_License)* https://github.com/riouxr/Set-Driven-Key	**Easy Drivers** *(GPL-3.0_License)* https://extensions.blender.org/add-ons/easy-driver/

FreeMoCap-v2.0 *(AGPL-3.0-License)* https://github.com/freemocap/freemocap	**Deadface-v1.0** *(MIT Licence)* https://github.com/Qaanaaq/DeadFace
BlendArMocap (FreeMoCapRig to "Rigify") *(GPL-3.0)* https://github.com/cgtinker/BlendArMocap	**Faceit** *(GPL-3.0-License)* https://superhivemarket.com/products/faceit

X-PosePicker/GPL-Picker (R-F) *(GPL-3.0-License)* https://superhivemarket.com/products/x-pose-picker	**Selector_RigMesh** https://github.com/WillyamBradberry/Blender **Selector_RigBone** https://phihung250693.gumroad.com/l/iyutsk
Animation-Layers *(GPL-3.0-License)* https://superhivemarket.com/products/animation-layers	**F-Curve-Wizard** *(GPL-3.0-License)* https://superhivemarket.com/products/f-curve-wizrd
RigUI *(GPL-3.0-License)* https://superhivemarket.com/products/rig-ui	**Dynamic-Parent** *(GPL-3.0-License)* https://github.com/romanvolodin/dynamic_parent
ReTime *(GPL-3.0-License)* https://superhivemarket.com/products/retime	**DJV** (video-FrameNumber) *(BSD-3-Clause-Licence)* https://github.com/darbyjohnston/DJV
Onion-Skin-Tools *(GPL-3.0-License)* https://superhivemarket.com/products/onion-skin-tools	**Playblast** (preview animation) *(GPL-3.0-License)* https://github.com/RxLaboratory/DuBlast

Lazy-composer-2-pro *(GPL-3.0-License)* https://superhivemarket.com/products/lazy-composer-2-pro	**Render-Statistics** *(GPL-3.0-License)* https://superhivemarket.com/products/render-stats
VFX BreakdownMaker (GPL-3.0-License) https://superhivemarket.com/products/breakdown-maker	**openTIMELINE** (.otio) *(GPL-3.0-License)* https://studio.blender.org/blog/opentimelineio-in-blender/

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Useful Links *(Compatible with "Blender-v5.0" for FX)*
Freezing Effect Generator (Frost) *(GPL-3.0-License)* https://maroc777772.gumroad.com/l/Freezing-Effect-Generator
Alt Tab Easy Fog (Fog) *(GPL-3.0-License)* https://superhivemarket.com/products/alt-tab-easy-fog
Water Shader (Water) *(GPL-3.0-License)* https://chuckcg.gumroad.com/l/sxbcnw https://superhivemarket.com/products/flipfluids https://superhivemarket.com/products/stylized-water
Dynamic Rain (Rain) *(GPL-3.0-License)* https://cgcool.gumroad.com/l/lxmii
Particle-X (Particles) *(GPL-3.0-License)* https://superhivemarket.com/products/particles-x
Electro (Lighting) *(GPL-3.0-License)* https://maroc777772.gumroad.com/l/yupnu
Dust Particles (Dust) (*Royalty-Free*) https://superhivemarket.com/products/dust-particles
Smoke Scatter (Smoke) *(GPL-3.0-License)* https://cgcool.gumroad.com/l/ybsfu
Fire Scatter (Fire) *(GPL-3.0-License)* https://cgcool.gumroad.com/l/xrwyo
Object Dropper (Gravity) *(GPL-3.0-License)* https://superhivemarket.com/products/physics-dropper https://superhivemarket.com/products/auto-smear-frames
Open Scatter (Object Scatter) *(GPL-3.0-License)* https://superhivemarket.com/products/openscatter
X-muscle-system *(GPL-3.0-License)* https://superhivemarket.com/products/x-muscle-system

#Note, generative a.i. video models are capable of producing effects.
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Software Website (Github)	License
7Zip (File Archiver) https://github.com/ip7z/7zip	*(LGPL, BSD 3-clause*
Aegisub (Subtitle Editor) https://github.com/Aegisub-Editor	*MIT license*
Audacity (Audio Editor) https://github.com/audacity/audacity	*GPL-3.0 license*
Balena Etcher (Electronics) https://github.com/balena-io/etcher	*Apache-2.0 license*
BeeRef (Image Reference Projection) https://github.com/rbreu/beeref	*GPL-3.0 license*
Bitcoin-Core (Wallet/Node) https://bitcoin.org/en/bitcoin-core/	*MIT license*
Blender (3D Visual Creativity Suite) https://github.com/blender/blender	*GPL-3.0 license*
Comfy UI (Image Generation) https://github.com/comfyanonymous/ComfyUI	*GPL-3.0 license*
Docker (Software Virtualization) https://github.com/docker/docker-install	*Apache-2.0 license*
DJV (FrameNumber) https://github.com/darbyjohnston/DJV	*BSD-3 Clause Licence*
Electrum (BTC Wallet) https://github.com/spesmilo/electrum	*MIT license*
Deadface (Motion Capture) https://github.com/Qaanaaq/DeadFace	*MIT license*
Flameshot (Reference Image Capture) https://github.com/flameshot-org/flameshot	*GPL-3.0 license*
FreeCAD (CAD Engineering) https://github.com/FreeCAD/FreeCAD	*LGPL license*
FreeMoCap (Motion Capture) https://github.com/freemocap/freemocap	*AGPL-3.0 license*
Firefox (Web Private Browser) [faster] https://github.com/mozilla/	*MPL-2.0 license*
GIMP (Image Manipulation) https://github.com/GNOME/gimp	*GPL-3.0 license*
Godot (Game Design Engine) https://github.com/godotengine/godot	*MIT license*
Folder-Structure (Version Control) https://github.com/Vidhey012/Folder-Structure-Visualizer	*GPL-2.0-license*
HandBrake (Video Transcoder) https://github.com/HandBrake/HandBrake	*GPL-2.0 license*
Meshroom (panarama) https://github.com/alicevision/Meshroom	*Mozilla Public v-2.0*
Inkscape (Image Graphics [Vector]) https://github.com/inkscape/inkscape	*GPL-2.0 license*
Kinovea (SyncVideo) https://github.com/Kinovea/Kinovea	*GPL-2.0 license*
Krita (2D Visual Creativity Suite) https://github.com/KDE/krita	*GPL-3.0 license*

LibreCAD (Drafter Architecture) https://github.com/LibreCAD/LibreCAD	*GPL-2.0 license*
LibreOffice (Productivity Office Suite) https://github.com/libreoffice	*GPL-2.0 license*
LMMS (Music Design) https://github.com/LMMS/lmms	*GPL-2.0 license*
Ollama (Run [llm]) https://github.com/ollama/ollama	*MIT license*
OpenCode (Agentic-Code) https://github.com/anomalyco/opencode/	*MIT license*
OBS Studio (Broadcasting) https://github.com/obsproject/obs-studio	*GPL-2.0 license*
real-ESRGAN (video-Upscaler-4k) https://github.com/xinntao/Real-ESRGAN	*BSD-3 Clause*
Pinokio (one-click install AI Models) https://github.com/pinokiocomputer/pinokio	*MIT license*
Proton VPN (Virtual Private Network) https://github.com/ProtonVPN/win-app	*GPL-3.0 license*
PyCharm CE (Python IDE) https://github.com/phracek/pycharm-community-edition	*GPL-2.0 license*
qBittorrent (File Sharing) https://github.com/qbittorrent/qBittorrent	*GPL-2.0 license*
QEMU (Emulator) https://github.com/qemu/qemu	*GPL-2.0 license*
Fstl (File Data Viewer) https://github.com/fstl-app/fstl	*MIT license*
Musescore (Music Notation) https://github.com/musescore/MuseScore	*GPL-3.0 license*
Natron (2D Compositing)(VFX) https://github.com/NatronGitHub/Natron	*GPL-2.0 license*
Neural Note (audio-to-midi) https://github.com/DamRsn/NeuralNote	*Apache-2.0 license*
Storyboarder (storyboard/animatics) https://github.com/wonderunit/storyboarder	*Custom license*
SumatraPDF (PDF, EPUB,CRB Reader) https://github.com/sumatrapdfreader/sumatrapdf	*GPL-3.0 license*
TOR Browser (Private Browsing) [slow] https://github.com/TheTorProject/gettorbrowser	*unknown*
Trelby (Screenplay Software) https://github.com/trelby/trelby	*GPL-2.0 license*
Ultimaker Cura (3D Printing) https://github.com/Ultimaker/Cura	*LGPL 3.0 license*
Ultimate Vocal Remover V5 (Split) https://github.com/Anjok07/ultimatevocalremovergui	*MIT license*
VLC Media Player (Media Player) https://github.com/videolan/vlc	*mixed*
VS Code (Code IDE) https://github.com/microsoft/vscode	*MIT license*
WineHQ (Productivity Office Suite) https://wiki.winehq.org/Download	*LGPL*

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"The Ten Essential Roles On The New Pipeline"
Producer A producer guides a film from its beginning to its completion. They have a hand in organizing and scheduling, budgeting and hiring, creative problem-solving and overseeing, and marketing and distributing. A producer may be a self-employed contractor, or subject to the authority of an employer such as a production company or studio. They are involved throughout all phases of production from inception to completion. <citation>
Attorney-At-Law [Lawyer] A lawyer (also called attorney, counsel, or counselor) is a licensed professional who advises and represents others in legal matters. <citation>
Accountant An accountant is a financial professional who reviews and analyses financial records and keeps track of a company's or individual's income, expenditures, and liabilities. An accountant may also work in project planning, cost analysis, auditing, and financial decision-making. Some specialize in tax preparation and tax planning. Accountants may work for large companies or external accounting firms. They must meet state-specific educational and testing requirements and are certified by national professional associations. <citation>
Director This person is responsible for designing sets, overseeing construction workers and other artists, and playing a part in figuring out the overall aesthetic of a movie production. <citation>.
Storyboard-Artist / Layout-Artist / Animatic-Artist This person creates animatics, which are sequences of images, shots, or sketches that are used to plan a video. Animatics are used in many fields, including animation, television commercials, and movie production. Animatics are a technique that comes after storyboarding, and they can help ensure that a project is on track and will be effective. They can be used to: see how the final product might look. Give a rough draft of how a particular idea will play out. Animatics are usually made by editing storyboard images together with dialogue, sound effects, and music. <citation>
Technical Artist (Rigging and Programming) This person (Programmer) helps video game development teams create interactive, visually appealing games for consoles and apps. They use both artistic and coding skills to integrate artwork and animation into complex game systems and film. <citation>.
Artificial Intelligence Operator [Story] [Concept] [Audio] [Mesh] [Look] [Movement] This (Developer) is a professional who designs, trains, and monitors AI systems. They work as a liaison between human operators and AI systems, ensuring that AI systems are integrated into existing workflows. <citation>.
VFX Compositing Artist This is the last piece of the puzzle you need to make effects look realistic. It combines the work of animators, videographers, and special effects artists to create effects that blur lines. <citation>.
Video Editor This person uses scenes, takes, and shots to create a cohesive story for the screen. Editors use continuity editing, cutaways, and transitions to evoke certain emotions from the viewer and properly execute an entertaining plot. Video editors for film cut a scene from different angles, which directs the viewer to certain details in a story. <citation>.
Graphic Designer Graphic designers create visual concepts, using computer software or by hand, to communicate ideas that inspire, inform, and captivate consumers. <citation>

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Budget

Category	Item	Projected Cost (JMD)	Actual Budget (JMD)	Difference (JMD)
Land Acquisition	Land Purchase	3,500,000.00		
Core Construction & Infrastructure	(20ft / 40ft) Shipping Container	1,000,000.00		
Core Construction & Infrastructure	Pit	35,000.00		
Core Construction & Infrastructure	Septic Tank	35,000.00		
Core Construction & Infrastructure	Manhole	35,000.00		
Core Construction & Infrastructure	Water Tank	35,000.00		
Building Materials	Gravel, Marl, Cement, Sand	500,000.00		
Plumbing & Fixtures	Toilet	15,000.00		
Plumbing & Fixtures	Bath Fixtures	15,000.00		
Plumbing & Fixtures	Pipes	15,000.00		
Plumbing & Fixtures	Sink (metal)	15,000.00		
Plumbing & Fixtures	Sink (plastic)	15,000.00		
Utilities & Energy	Solar Generator	250,000.00		
Utilities & Energy	Starlink/Data Plan	5,500.00		
Appliances & Furniture	Gas Stove	10,000.00		
Appliances & Furniture	Fridge	30,000.00		
Appliances & Furniture	Microwave	25,000.00		
Appliances & Furniture	Washing Machine	35,000.00		
Appliances & Furniture	Bed	50,000.00		
Appliances & Furniture	Table	15,000.00		
Appliances & Furniture	Air conditioning unit	160,000.00		
Appliances & Furniture	Dining set	160,000.00		
Appliances & Furniture	Sofa	75,000.00		
Technology & Electronics	Phone	75,000.00		
Technology & Electronics	Graphics Tablet	75,000.00		
Technology & Electronics	Laptop	200,000.00		
Technology & Electronics	e-Mouse	20,000.00		
Technology & Electronics	e-Pencil	20,000.00		
Technology & Electronics	Stereo Set	20,000.00		
Technology & Electronics	Solar Generator (additional)	30,000.00		
Power Tools	Cordless Drill	35,000.00		
Power Tools	Cordless Angle-grinder	35,000.00		
Power Tools	Portable Welding-plant	35,000.00		
Power Tools	Cordless Circle saw	35,000.00		
Power Tools	Cordless Chainsaw	35,000.00		
Workshop Tools	Hammer	3,500.00		
Workshop Tools	Pipe-cutter	3,500.00		
Workshop Tools	Level	3,500.00		
Workshop Tools	Tape measure	3,500.00		
Workshop Tools	Hacksaw	3,500.00		
Workshop Tools	Saw	3,500.00		
Workshop Tools	Wire-cutter	3,500.00		
Workshop Tools	Adjustable Wrench	3,500.00		
Workshop Tools	Tile-Cutter	3,500.00		
Workshop Tools	Shovel (big)	3,500.00		
Workshop Tools	Ladder	3,500.00		
Workshop Tools	Wheel-Borrow	3,500.00		
Workshop Tools	Cement bucket	1,600.00		
Workshop Tools	Plastic bucket	1,600.00		
Workshop Tools	Mallet	1,600.00		
Painting Supplies	Paint roller & tray bundle	4,800.00		
Painting Supplies	Paint Brushes (x3)	4,800.00		

Budget Estimate Summary

Category	Projected Cost (JMD)	Actual Budget (JMD)	Difference (JMD)
Appliances & Furniture	560,000.00		
Building Materials	500,000.00		
Core Construction & Infrastructure	1,140,000.00		
Land Acquisition	3,500,000.00		
Painting Supplies	9,600.00		
Plumbing & Fixtures	75,000.00		
Power Tools	175,000.00		
Technology & Electronics	440,000.00		
Utilities & Energy	255,000.00		
Workshop Tools	46,800.00		
Total (JMD)	6,701,900.00		
Total (USD)	$41,886.88		

Exchange Rate Applied: 1 USD = 160 JMD

Figure 1: Table showing a budget for the studio.

*Labour cost not included. *Adhesive (weld-rods, nails, tin-set, paint, ect.) cost not included. *Also, gas-generator, solar-light, windows, and safety-gear are not included.

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